Analysis of Boolean Functions Solution Manual

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Preface

These solutions are the culmination of our collective reading of Ryan O'Donnell's *Analysis of Boolean Functions* over the Spring 2023 semester. The hope is for this to be used as a reference for students learning the topic for the first time just like us.

Contents

1 Boolean Functions and the Fourier Expansion

Problem 1.2 Recall that the set of boolean functions $\{f | f : \{-1,1\}^n \to \{-1,1\}\}$ is a vector space of dimension 2^n with parity functions χ_S (for $S \subseteq [n]$) being the basis vectors. Let $\widehat{f}(S)$ be the only non-zero Fourier coefficient. Since the range of f is $\{-1,1\}$, $\widehat{f}(S)$ must be either -1 or 1. By symmetry then, there are $2^n \cdot 2 = 2^{n+1}$ functions with exactly one non-zero Fourier coefficient. $\square$

Problem 1.3 Let $A = f^{-1}(\{1\}) \subset \{-1,1\}^n$. We are given that $|A|$ is odd and $n \geq 2$. Let $S \subseteq [n]$. Then, we have $\widehat{f}(S) = \mathbb{E}_x[f(x)\cdot\chi_S(x)] = 2\cdot\Pr_x[f(x) = \chi_S(x)] - 1$. To show that $\widehat{f}(S) \neq 0$, then, it suffices to show that $\Pr_x[f(x) = \chi_S(x)] \neq \frac{1}{2}$. Let $B = \chi_S^{-1}(\{1\})$. Observe that $|B| = \begin{cases} 2^{n-1} & \text{if } S \neq \varnothing \\ 2^n & \text{if } S = \varnothing \end{cases}$. Since $n \geq 2$, $|B|$ is even in either case.

Then, we have:

$$\begin{aligned}
\Pr_x[f(x) = \chi_S(x)] &= \Pr_x[f(x) = \chi_S(x) = 1] + \Pr_x[f(x) = \chi_S(x) = -1] \\
&= \frac{|A \cap B|}{2^n} + \frac{|A^c \cap B^c|}{2^n} \\
&= \frac{|A \cap B|}{2^n} + \frac{2^n - |A \cup B|}{2^n} \\
&= \frac{|A \cap B|}{2^n} + \frac{2^n - |A| - |B| + |A \cap B|}{2^n} \\
&= \frac{2|A \cap B| + 2^n - (|A| + |B|)}{2^n}.
\end{aligned}$$

Note that the numerator is odd because $|A| + |B|$ is odd. In particular, the numerator cannot be 2^{n-1}, so $\Pr_x[f(x) = \chi_S(x)] \neq \frac{1}{2}$, which means $\widehat{f}(S) \neq 0$ as desired. $\square$

Problem 1.4

$$\begin{aligned}
\mathbb{E}_y[f(y)] &= \mathbb{E}_y[\sum_{S\subseteq[n]} \widehat{f}(S) y^S] = \sum_{S\subseteq[n]} \widehat{f}(S)\cdot\mathbb{E}_y[y^S] && \text{(Linearity of Expectation)} \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)\cdot\mathbb{E}_y[\Pi_{i\in S} y_i] = \sum_{S\subseteq[n]} \widehat{f}(S)\cdot\Pi_{i\in S}\mathbb{E}_y[y_i] && \text{(Independence)} \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)\cdot\Pi_{i\in S}\mu_i = F(\mu)
\end{aligned}$$

$\square$

Problem 1.5

(i) Suppose for the sake of contradiction that $|\widehat{f}(S)|, |\widehat{f}(T)| > \frac{1}{2}$ for $S \neq T$. Then $1 - 2\cdot\operatorname{dist}(f, \chi_S) > \frac{1}{2}$ and $1 - 2\cdot\operatorname{dist}(f, \chi_T) > \frac{1}{2}$ by definition, which yields $\operatorname{dist}(f, \chi_S), \operatorname{dist}(f, \chi_T) < \frac{1}{4}$. But then:

$$\begin{aligned}
\operatorname{dist}(\chi_S, \chi_T) &\leq \operatorname{dist}(f, \chi_S) + \operatorname{dist}(f, \chi_T) \\
&< \frac{1}{4} + \frac{1}{4} \\
&= \frac{1}{2}.
\end{aligned}$$

This is a contradiction, since $\operatorname{dist}(\chi_S, \chi_T) = \frac{1}{2}$ (as $\langle \chi_S, \chi_T\rangle = 0$). Hence, no such S, T can exist, as desired.

(ii) Consider $f : \{-1,1\}^n \to \mathbb{R}$ given by $f(x) = \frac{3}{5}\Pi_{i=1}^n x_i + \frac{4}{5}$

$\square$

Problem 1.7 Fix $S \subseteq [n]$. Then we have:

$$\begin{aligned}
\mathbb{E}_f[\widehat{f}(S)] &= 2^{-n}\sum_f \widehat{f}(S) \\
&= 2^{-n}\sum_f \langle f, \chi_S\rangle \\
&= 2^{-n}\langle \sum_f f, \chi_S\rangle && \text{(bilinearity)} \\
&= 2^{-n}\langle 0, \chi_S\rangle && \text{(can partition function space into } (f,-f) \text{ pairs)} \\
&= 0.
\end{aligned}$$

To clarify the above logic, let A be the set of functions $f\colon \{-1,1\}^n \to \{-1,1\}$ for which $f(1,1,\ldots,1) = 1$ and B be the set where $f(1,1,\ldots,1) = -1$. A and B partition the function space and have the property that $f \in A \iff -f \in B$. Therefore $\sum_f f = \sum_A f + \sum_B f = 0$.

We now compute the variance:

$$\begin{aligned}
\mathbf{Var}_f[\widehat{f}(S)] &= \mathbb{E}_f[\widehat{f}(S)^2] - \mathbb{E}_f[\widehat{f}(S)]^2 \\
&= \mathbb{E}_f[\widehat{f}(S)^2] \\
&= \mathbb{E}_f[\langle f, \chi_S\rangle^2] \\
&= \mathbb{E}_f[\langle f\cdot\chi_S, \chi_S\rangle^2] && \text{(permutation of function space)} \\
&= \mathbb{E}_f[\mathbb{E}_x[f(x)\cdot\chi_S(x)^2] \\
&= \mathbb{E}_f[\langle f, 1\rangle^2] && (\chi_S^2 = 1) \\
&= \mathbb{E}_f[\widehat{f}(\varnothing)].
\end{aligned}$$

To clarify the above logic, observe that χ_S is fixed and $\{f\} = \{f\cdot\chi_S\}$, so replacing f with χ_S does not change the expectation. But note that there is no dependence on S. Therefore $\mathbb{E}_f[\widehat{f}(S)^2] = \mathbb{E}_f[\widehat{f}(\varnothing)^2]$ for all S. Finally, by Parseval's, we have:

$$\begin{aligned}
1 &= \mathbb{E}_f\left[\sum_T \widehat{f}(T)^2\right] \\
&= \sum_T \mathbb{E}_f\left[\widehat{f}(T)^2\right] \\
&= \sum_T \mathbb{E}_f[\widehat{f}(\varnothing)^2] \\
&= 2^n\cdot\mathbb{E}_f[\widehat{f}(\varnothing)^2].
\end{aligned}$$

We conclude that $\mathbb{E}_f[\widehat{f}(S)^2] = E_f[\widehat{f}(\varnothing)] = 2^{-n}$, as desired. $\square$

Problem 1.8

(a)

$$\widehat{f^\dagger}(S) = \langle f^\dagger(x), \chi_S(x)\rangle = \langle -f(-x), \chi_S(x)\rangle = -\frac{1}{2^n}\sum_{x\in\{-1,1\}^n} f(-x)\chi_S(x)$$

Note that

$$\chi_S(x) = \begin{cases} -\chi_S(-x) \text{ if } |S| \bmod 2 = 1 \\ \chi_S(-x) \text{ if } |S| \bmod 2 = 0 \end{cases}$$

Then,

$$\widehat{f^\dagger}(S) = \begin{cases} \frac{1}{2^n}\sum_{x\in\{-1,1\}^n} f(-x)\chi_S(-x) \text{ if } |S| \bmod 2 = 1 \\ -\frac{1}{2^n}\sum_{x\in\{-1,1\}^n} f(-x)\chi_S(-x) \text{ if } |S| \bmod 2 = 0 \end{cases} = \begin{cases} \widehat{f}(S) \text{ if } |S| \bmod 2 = 1 \\ -\widehat{f}(S) \text{ if } |S| \bmod 2 = 0 \end{cases}$$

(b) $f^{\text{odd}}(x) + f^{\text{even}}(x) = \frac{f(x)-f(-x)}{2} + \frac{f(x)+f(-x)}{2} = f(x)$. If f is odd, then $f(-x) = -f(x)$ and $f^{\text{odd}}(x) = \frac{f(x)+f(x)}{2} = f(x)$. Similarly, if $f(x) = f^{\text{odd}}(x)$ then $f(x) = -f(-x)$ which means f is odd. A correspondingly similar argument can be used to show the case for even f.

(c) Note that

$$\begin{aligned}
f &= f^{\text{odd}} + f^{\text{even}} \\
f^\dagger &= f^{\text{odd}} - f^{\text{even}}
\end{aligned}$$

Thus,

$$f + f^\dagger = 2f^{\text{odd}} = \sum_{S\subseteq[n]} \widehat{f}(S)\chi_S + \sum_{\substack{S\subseteq[n] \\ |S| \text{ odd}}} \widehat{f}(S)\chi_S - \sum_{\substack{S\subseteq[n] \\ |S| \text{ even}}} \widehat{f}(S)\chi_S \quad = 2\sum_{\substack{S\subseteq[n] \\ |S| \text{ odd}}} \widehat{f}(S)\chi_S$$

Which means $f^{\text{odd}} = \sum_{\substack{S\subseteq[n] \\ |S| \text{ odd}}} \widehat{f}(S)\chi_S$. Then,

$$f^{\text{even}} = f - f^{\text{odd}} = \sum_{S\subseteq[n]} \widehat{f}(S)\chi_S - \sum_{\substack{S\subseteq[n] \\ |S| \text{ odd}}} \widehat{f}(S)\chi_S = \sum_{\substack{S\subseteq[n] \\ |S| \text{ even}}} \widehat{f}(S)\chi_S$$

$\square$

Problem 1.9

(a) We will use the notation $\{0,1\}$ rather than $\{\mathrm{F}, \mathrm{T}\}$. Let $f\colon \{0,1\}^n \to \{0,1\}$ be arbitrary. For each $a \in \{0,1\}^n$, let $\mathbf{1}_a(x) = \prod_{i=1}^n (1 - a_i - x_i + 2a_i x_i)$. By construction, $\mathbf{1}_a(x) = \begin{cases} 1 & \text{if } a = x \\ 0 & \text{if } a \neq x \end{cases}$. Moreover, $\mathbf{1}_a(x)$ is multilinear. Therefore, $f(x) = \sum_{a\in\{0,1\}^n} f(a)\cdot\mathbf{1}_a(x)$ is a multilinear representation for f.

(b) It is enough to show that the zero function has no nonzero multilinear representation $q(x) = \sum_{S\subseteq[n]} c_S x^S$ (if some function f had distinct representations p, p' then consider $q = p - p'$). Suppose for contradiction such a q exists. Let $T \subseteq [n]$ be minimal with $c_T \neq 0$, so that $S \subsetneq T$ does not hold for any S where $c_S \neq 0$. Let $a \in \{0,1\}^n$ be given by $a_i = \begin{cases} 1 & \text{if } i \in T \\ 0 & \text{if } i \notin T \end{cases}$. Then:

$$q(a) = \sum_{S\subseteq[n]} c_S x^S = \sum_{S\subseteq T} c_S x^S = c_S \neq 0.$$

This is a contradiction, meaning no such q exists, which proves uniqueness of multilinear representation.

(c) Let $\{c_S\}$ be the coefficients for the multilinear representation of $f\colon \{0,1\}^n \to \{0,1\}$ .Let $T \subseteq [n]$ be arbitrary. We aim to show that $c_T \in [-2^n, 2^n] \cap \mathbb{Z}$. We make the following claim:

$$\sum_{S\subseteq T}(-1)^{|S|}\sum_{R\subseteq S} c_R = (-1)^{|T|}c_T$$

To show this, take any $R \subsetneq T$ and consider the coefficient of c_R in the sum:

$$\sum_{S:\ R\subseteq S\subseteq T}(-1)^{|S|} = (1-1)^{|T|-|R|} = 0.$$

Meanwhile, c_T only shows up once in the sum with coefficient $(-1)^{|T|}c_T$. This proves our claim.

Now, observe that for a set $S \subseteq [n]$, we have $\sum_{R\subseteq S} c_R = f(a_S)$ where $(a_S)_i := \begin{cases}1 & \text{if } i \in S\\ 0 & \text{if } i \notin S\end{cases}$.

Therefore, we have:

$$|c_T| = \left|\sum_{S\subseteq T}(-1)^{|S|}\cdot f(a_S)\right|.$$

Finally, recall that $f(a_S) \in \{0,1\}$, which means that $c_T \in \mathbb{Z}$, and which also gives the following bounds:

$$|c_T| \le \max\left\{\sum_{\substack{S\subseteq T\\ |S|\text{ even}}} 1,\ \sum_{\substack{S\subseteq T\\ |S|\text{ odd}}} 1\right\} = \begin{cases}2^{|T|-1} & \text{if } T \neq \varnothing\\ 1 & \text{if } T = \varnothing\end{cases}.$$

To clarify, we could make c_T as positive as possible by setting $f(a_S) = 1$ for all even $|S|$, and $f(a_S) = 0$ for all odd $|S|$. Or we could make c_T as negative as possible with the opposite setting. Both cases give us the sum of every other binomial coefficient $\binom{|T|}{k}$, which is $2^{|T|-1} \le 2^{n-1}$ when $T \neq \varnothing$. Therefore, we have actually shown a slightly stronger statement, that $c_T \in [-2^{n-1}, 2^{n-1}] \cap \mathbb{Z}$ whenever $n \ge 1$ (and $c_\varnothing \in \{0,1\}$ always).

(d) Take $f\colon \{0,1\}^n \to \{0,1\}$ with Fourier expansion p and $\{0,1\}$-Fourier expansion q. If $x \in \{0,1\}^n$ then we have $\frac{1}{2} - \frac{1}{2}p(1-2x_1,\dots,1-2x_n) = \frac{1}{2} - \frac{1}{2}\cdot(-1)^{f(x)} = f(x)$. By uniqueness of multilinear representation, it follows that $q(x) = \frac{1}{2} - \frac{1}{2}p(1-2x_1,\dots,1-2x_n)$ as desired.

□

Problem 1.10

(a) Suppose $a, b \in \mathbb{R}$ with $b \neq 0$ and $a + bf \neq 0$. Write $f(x) = \sum_{S\subseteq[n]} \widehat{f}(S)x^S$ so that $(a+bf)(x) = a + \sum_{S\subseteq[n]} b\widehat{f}(S)x^S$. If $S \neq \varnothing$ then clearly $\widehat{(a+bf)}(S) = b\widehat{f}(S) = 0 \iff \widehat{f}(s) = 0$ since $b \neq 0$. Likewise, $\widehat{(a+bf)}(\varnothing) = a + b\widehat{f}(\varnothing) = 0$ only if $\deg f \neq 0$, by our assumption that $a + bf \neq 0$. And so either $\deg f = 0$ in which case $\deg(a+bf) = \deg f = 0$ as well, or $\deg f \neq 0$ in which case $\deg(a+bf) = \deg f$, which is what we wanted to show.

(b) ($\Rightarrow$) Suppose $\deg f \le k$. By definition, we can write:

$$f(x) = \sum_{\substack{S\subseteq[n]\\ |S|\le k}} \widehat{f}(S)x^S.$$

If we take $g_S := x^S$ then each g_S depends on at most k input coordinates, meaning f is of the desired form.

($\Leftarrow$) Suppose $f = \alpha_1 g_1 + \cdots + \alpha_m g_m$ where $\alpha_i \in \mathbb{R}$ and g_i depend on at most k input coordinates. For a given g_i, let $x_{i_1}, \dots, x_{i_k}$ be the input coordinates g_i depends on. Then for $a \in \{-1,1\}^k$ we can write the following indicators

$$\mathbf{1}_a(x_{i_1},\dots,x_{i_k}) = \prod_{j=1}^{k}\frac{1+a_j x_{i_j}}{2} = \begin{cases}1 & \text{if } a_j = x_{i_j}\forall j\\ 0 & \text{otherwise}\end{cases}$$

we can write $g_i(x) = \sum_{a\in\{-1,1\}^k}\mathbf{1}_a(x_{i_1},\dots,x_{i_k})$. But $\deg(g_i) \le k$ so by applying part (a) finitely many times we conclude $\deg(f) = \deg(\alpha_1 g_1 + \cdots + \alpha_m g_m) \le k$.

(c) Omitting this part because it is tedious and uninteresting.

□

Problem 1.12

(a) We can show this by induction. For $n = 1$ we have $H_{2^1}[\gamma, x] = \begin{cases}-1 & \text{if } \gamma = x = 1\\ 1 & \text{otherwise}\end{cases} = (-1)^{\gamma\cdot x}$.

Now, supposing $H_{2^n}[\gamma, x] = (-1)^{\gamma\cdot x}$ for a specific $n \in \mathbb{N}$, we aim to show the same for $n+1$. If $\gamma, x \in \mathbb{F}_2^{n+1}$ observe that we can write $\gamma = \gamma'||a$ and $x = x'||b$ (concatenation) where $\gamma', x' \in \mathbb{F}_2^n$ and $a, b \in \mathbb{F}_2$. Since a, b are the most significant bits of γ and x, respectively, we can say $H_{2^{n+1}}[\gamma, x] = \begin{cases}-H_{2^n}[\gamma', x'] & \text{if } a = b = 1\\ H_{2^n}[\gamma', x'] & \text{otherwise}\end{cases}$. But this is simply equal to $H[\gamma', x']\cdot(-1)^{a\cdot b}$ which by the inductive hypothesis is $(-1)^{\gamma'\cdot x'}\cdot(-1)^{a\cdot b} = (-1)^{\gamma'\cdot x' + a\cdot b} = (-1)^{\gamma\cdot x}$, as desired.

(b) Let $f\colon \mathbb{F}_2 \to \mathbb{R}$. Let $i \in \mathbb{F}_2^n$ be associated with $S_i \subset \{0,\dots,n-1\}$ according to the indexing scheme mentioned in the question. Then we have:

$$\begin{aligned}H_{2^n}f[i] &= H_{2^n}[i]\cdot f\\ &= \sum_{x=0}^{2^n-1}(-1)^{i\cdot x}\cdot f(x).\end{aligned}$$

On the other hand, we have:

$$\begin{aligned}2^n\widehat{f}(S_i) &= 2^n\mathbb{E}_x[f(x)\cdot\chi_{S_i}(x)]\\ &= \sum_{x=0}^{2^n-1} f(x)\cdot(-1)^{\sum_{j\in S_i}x_j}\\ &= \sum_{x=0}^{2^n-1} f(x)\cdot\prod_{\ell=1}^{n}(-1)\cdot\mathbf{1}[x_\ell = -1]\cdot\mathbf{1}[\ell\in S_i]\\ &= \sum_{x=0}^{2^n-1} f(x)\cdot(-1)^{i\cdot x}.\end{aligned}$$

Hence $2^{-n}H_{2^n}f = \widehat{f}$.

(c) Let $f \in \mathbb{R}^{2^n}$. We can express $f = f_1||f_2$ where $f_1, f_2 \in \mathbb{R}^{2^{n-1}}$. Then, consider the following algorithm ALG(n, f) for computing $H_{2^n}f$:

- If $n = 0$: return f.
- If $n \ge 1$: return $\begin{bmatrix}\text{ALG}(n-1, f_1+f_2)\\ \text{ALG}(n-1, f_1-f_2)\end{bmatrix}$.

This algorithm is correct because for $n \ge 1$ we have:

$$\begin{aligned}H_{2^n}f &= \begin{bmatrix}H_{2^{n-1}} & H_{2^{n-1}}\\ H_{2^{n-1}} & -H_{2^{n-1}}\end{bmatrix}\begin{bmatrix}f_1\\ f_2\end{bmatrix}\\ &= \begin{bmatrix}H_{2^{n-1}}(f_1+f_2)\\ H_{2^{n-1}}(f_1-f_2)\end{bmatrix}.\end{aligned}$$

We now evaluate its complexity. Its recursion tree has depth n, and the recursive step at depth k takes $2\cdot 2^{n-k-1} = 2^{n-k}$ additions/subtractions (where $0 \le k \le n-1$), because one needs to compute $f_1+f_2, f_1-f_2 \in \mathbb{R}^{2^{n-k-1}}$. No computation needs to be done at the depth n level. Hence, the total complexity is given by:

$$\begin{aligned}\sum_{k=0}^{n-1}2^{n-k}\cdot 2^k &= \sum_{k=0}^{n-1}2^n\\ &= n2^n.\end{aligned}$$

Hence, we have found an algorithm for computing $H_{2^n}f$ using only $n2^n$ additions/subtractions, as desired.

(d) Since H_{2^n} is orthogonal and each row vector has magnitude $\sqrt{2^n}$, we have $H_{2^n}^2 = 2^n\cdot I_{2^n}$. Using part (b), we have $\widehat{\widehat{f}} = 2^{-n}H_{2^n}\widehat{f} = 2^{-2n}H_{2^n}^2 f = 2^{-n}f$, as desired.

□

Problem 1.13 Consider the convex function $\varphi(t) = t^{\frac{q}{p}}$ where $q > p$. By Jensen's inequality

$$\begin{aligned}\mathbb{E}_x[\varphi(f(x)^p)] &\ge \varphi(\mathbb{E}_x[f(x)^p])\\ \mathbb{E}_x[f(x)^q] &\ge \mathbb{E}_x[f(x)^p]^{q/p}\\ \mathbb{E}_x[f(x)^q]^{1/q} &\ge \mathbb{E}_x[f(x)^p]^{1/p}\\ \|f\|_q &\ge \|f\|_p\end{aligned}$$

To show the inequality holds for $q = \infty$ where $||f||_\infty := \max_{x\in\{-1,1\}^n}\{|f(x)|\}$, it suffices to show that for every f, $\lim_{q\to\infty}||f||_q = ||f||_\infty$. Suppose $M = ||f||_\infty$ so that $|f(x)| \le M$ for all $x \in \{-1,1\}^n$. Then, we have:

$$\begin{aligned}\lim_{q\to\infty}||f||_q &= \lim_{q\to\infty}\mathbb{E}_x\left[|f(x)|^q\right]^{1/q}\\ &\le \lim_{q\to\infty}\mathbb{E}_x\left[M^q\right]^{1/q}\\ &= M.\end{aligned}$$

And:

$$\begin{aligned}\lim_{q\to\infty}||f||_q &= \lim_{q\to\infty}\mathbb{E}_x\left[|f(x)|^q\right]^{1/q}\\ &= \lim_{q\to\infty}\frac{(\sum_x|f(x)|^q)^{1/q}}{2^{n/q}}\\ &\ge \lim_{q\to\infty}\frac{M}{2^{n/q}}\\ &= M && (\lim_{q\to\infty}2^{n/q} = 1)\end{aligned}$$

We conclude that $\lim_{q\to\infty}||f||_q = ||f||_\infty$ in general, so we are done. □

Problem 1.15 We denote $k = |K|$. Then,

$$\mathbb{E}[g(x)] = \frac{1}{2^{n-k}}\sum_{x\in\{0,1\}^{n-k}}g(x) = \frac{1}{2^{n-k}}\sum_{x\in\{0,1\}^{n-k}}\sum_{S\subseteq[n]}\widehat{f}(S)\chi_S(xz)$$

where xz denotes the complete n bit string including the fixed z bits. We split this sum into two parts

$$= \frac{1}{2^{n-k}}\sum_{x\in\{0,1\}^{n-k}}\sum_{T\subseteq K}\widehat{f}(T)z^T + \frac{1}{2^{n-k}}\sum_{x\in\{0,1\}^{n-k}}\sum_{S\not\subseteq K}\widehat{f}(S)\chi_S(xz)$$

Notice that $\sum_{T\subseteq K}\widehat{f}(T)z^T$ is a constant for every x since z is fixed and $\widehat{f}(T)$ is determined by f. Thus,

$$= \sum_{T\subseteq K}\widehat{f}(T)z^T + \frac{1}{2^{n-k}}\sum_{S\not\subseteq K}\sum_{x\in\{0,1\}^{n-k}}\widehat{f}(S)\chi_S(xz)$$

where we have also switched the order of summation in the second term. For any $S \not\subseteq K$, $\chi_S(xz)$ is a function of $m = |S| - |S \cap K| \geq 1$ unfixed bits. Note then that the number of $x \in \{0,1\}^{n-k}$ such that $\chi_S(xz)$ has odd parity is exactly equal to those such that $\chi_S(xz)$ has even parity. This is because there are $2^{n-k+m}\sum_{i=0}^{\lfloor \frac{m-1}{2} \rfloor}\binom{m}{2i}$ many x that result in one parity and $2^{n-k+m}\sum_{i=0}^{\lfloor \frac{m-1}{2} \rfloor}\binom{m}{2i+1}$ many that result in the other. Since the sum of the even binomial coefficients is equal to that of the odd, the claim follows.

Then, for each $S \not\subseteq K$, $\sum_{x\in\{0,1\}^{n-k}} \widehat{f}(S)\chi_S(xz) = 0$. Therefore,

$$\mathbb{E}[g(x)] = \sum_{T \subseteq K} \widehat{f}(T) z^T$$

□

Problem 1.16 By definition $\mathrm{dist}(f,1) = \mathbb{P}[f(x) \neq 1] = \mathbb{P}[f(x) = -1]$ and $\mathrm{dist}(f,-1) = \mathbb{P}[f(x) \neq -1] = \mathbb{P}[f(x) = 1]$. For $f : \{-1,1\}^n \to \{-1,1\}$

$$\begin{aligned}
\mathbf{Var}(f) &= 1 - \mathbb{E}[f^2] = 1 - (\mathbb{P}[f(x)=1] - \mathbb{P}[f(x)=-1])^2 \\
&= 1 - \mathbb{P}[f(x)=1]^2 - \mathbb{P}[f(x)=-1]^2 + 2\mathbb{P}[f(x)=1]\mathbb{P}[f(x)=-1] \\
&= (1+\mathbb{P}[f(x)=1])\mathbb{P}[f(x)=-1] - \mathbb{P}[f(x)=-1]^2 + 2\mathbb{P}[f(x)=1]\mathbb{P}[f(x)=-1] \\
&= \mathbb{P}[f(x)=-1](1+\mathbb{P}[f(x)=1] - \mathbb{P}[f(x)=-1]) + 2\mathbb{P}[f(x)=1]\mathbb{P}[f(x)=-1] \\
&= 4\mathbb{P}[f(x)=1]\mathbb{P}[f(x)=-1] = 4\,\mathrm{dist}(f,-1)\mathrm{dist}(f,1)
\end{aligned}$$

Notice that $\mathrm{dist}(f,1), \mathrm{dist}(f,-1) \leq 1$. Then,

$$\begin{aligned}
\mathbf{Var}[f] &= 4 \cdot \min(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) \cdot \max(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) \\
&= 4\epsilon \cdot \max(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) \leq 4\epsilon
\end{aligned}$$

Secondly, because $\mathrm{dist}(f,1) + \mathrm{dist}(f,-1) = 1$, it follows that $\max(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) \geq \frac{1}{2}$. Thus,

$$\begin{aligned}
\mathbf{Var}[f] &= 4 \cdot \min(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) \cdot \max(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) \\
&\geq 2 \cdot \min(\mathrm{dist}(f,1), \mathrm{dist}(f,-1)) = 2\epsilon
\end{aligned}$$

Therefore $2\epsilon \leq \mathbf{Var}[f] \leq 4\epsilon$. □

Problem 1.17 We denote $p = \mathbb{P}[f(x) = 1]$ and $q = \mathbb{P}[f(x) = -1]$. Then,

(1)
$$\mathbf{Var}[F] = \mathbb{E}[(F-\mu)^2] = \mathbb{E}[F^2] - \mu^2 \qquad \text{(Definition)}$$

(2)
$$\begin{aligned}
\mathbb{E}[(F-F')^2] &= \mathbb{E}[F^2] - 2\mathbb{E}[F'F] + \mathbb{E}[F'^2] && \text{(Linearity of Expectation)} \\
&= 2\mathbb{E}[F^2] - 2\mu^2 && \text{(Independence)} \\
&= 2\mathbb{E}[(F-\mu)^2]
\end{aligned}$$

(3)
$$\begin{aligned}
\mathbb{E}[(F-F')^2] &= 0 \cdot (p^2+q^2) + 4 \cdot (2pq) = 8pq \\
\mathbb{P}[F \neq F'] &= 2pq
\end{aligned}$$

Therefore,

$$\frac{1}{2}\mathbb{E}[(F-F')^2] = 2\mathbb{P}[F \neq F']$$

(4)
$$\begin{aligned}
\mathbb{E}[|F-\mu|] &= |1-\mu|p + |\mu+1|q \\
&= (1-\mu)p + (\mu+1)q \quad (-1 \leq \mu \leq 1 \implies |1-\mu|, |1+\mu| \geq 0) \\
&= (1-p+q)p + (p-q+1)q && (\mu = p-q) \\
&= p - p^2 + pq + pq - q^2 + q \\
&= 1 - (p^2+q^2) + 2pq && (p+q=1) \\
&= 4pq && ((p+q)^2 = 1)
\end{aligned}$$

Thus,

$$\mathbb{E}[|F-\mu|] = 2\mathbb{P}[F \neq F']$$

□

Problem 1.18 We perform a simple computation:

$$\begin{aligned}
\langle f^{=k}, f^{=\ell} \rangle &= \langle \sum_{S:\,|S|=k} \widehat{f}(S)\chi_S, \sum_{T:\,|T|=\ell} \widehat{f}(T)\chi_T \rangle \\
&= \sum_{S:\,|S|=k}\sum_{T:\,|T|=\ell} \widehat{f}(S)\widehat{f}(T)\langle \chi_S, \chi_T \rangle && \text{(bilinearity)} \\
&= \begin{cases} \sum_{S:\,|S|=k} \widehat{f}(S)^2 & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases} \\
&= \begin{cases} \mathbf{W}^k[f] & \text{if } k = \ell \\ 0 & \text{if } k \neq \ell \end{cases}.
\end{aligned}$$

This is what we wanted to show. □

Problem 1.19

(a) Suppose, for the sake of contradiction, that there exists an $f : \{-1,1\}^n \to \{-1,1\}$ such that $\mathbf{W}^1[f] = 1$ but is not of the form $f = \pm\chi_S$. This implies that $\widehat{f}(S_{i_1}), ..., \widehat{f}(S_{i_k}) \neq 0$ for some $1 < k \leq n$, where $1 \leq i_j \leq n$ and $|S_{i_j}| = 1$. Then, without loss of generality

$$f(\mathbf{1}) = \widehat{f}(S_{i_1}) + ... + \widehat{f}(S_{i_k}) = 1$$

Now denote $\mathbf{1_p} = (1,1,1,....,-1,...,1,1)$ where the pth index is set to -1. Then,

$$f(\mathbf{1}_{i_k}) = \widehat{f}(S_{i_1}) + ... - \widehat{f}(S_{i_k}) = \pm 1$$

If $f(\mathbf{1}_{i_k}) = 1$, then $\widehat{f}(S_{i_k}) = 0$ which contradicts our assumption that it is nonzero. If $f(\mathbf{1}_{i_k}) = -1$, then $\widehat{f}(S_{i_k}) = 1$ which by Parseval's means that $\widehat{f}(S_{i_1}), ..., \widehat{f}(S_{i_{k-1}}) = 0$ – contradiction.

(b) Suppose, for the sake of contradiction, that there exists an $f : \{-1,1\}^n \to \{-1,1\}$ such that $\mathbf{W}^{\leq 1}[f] = 1$ but depends on more than 1 input coordinate. If $\widehat{f}(\phi) = 0$ and $\widehat{f}(S_{i_1}), ..., \widehat{f}(S_{i_k}) \neq 0$ for some $1 < k \leq n$, where $1 \leq i_j \leq n$ and $|S_{i_j}| = 1$, then part (a) argues a contradiction. Otherwise, $\widehat{f}(\phi), \widehat{f}(S_{i_1}), ..., \widehat{f}(S_{i_k}) \neq 0$ for some $1 < k \leq n$, where $1 \leq i_j \leq n$ and $|S_{i_j}| = 1$. Then, without loss of generality

$$f(\mathbf{1}) = \widehat{f}(\phi) + \widehat{f}(S_{i_1}) + ... + \widehat{f}(S_{i_k}) = 1$$

Then,

$$f(\mathbf{1}_{i_k}) = \widehat{f}(\phi) + \widehat{f}(S_{i_1}) + ... - \widehat{f}(S_{i_k}) = \pm 1$$

If $f(\mathbf{1}_{i_k}) = 1$, then $\widehat{f}(S_{i_k}) = 0$ which contradicts our assumption that it is nonzero. If $f(\mathbf{1}_{i_k}) = -1$, then $\widehat{f}(S_{i_k}) = 1$ which by Parseval's means that $\widehat{f}(\phi), \widehat{f}(S_{i_1}), ..., \widehat{f}(S_{i_{k-1}}) = 0$ – contradiction.

(c) need to figure out

□

Problem 1.21

(a) Suppose, for the sake of contradiction, that there exists an $f : \{-1,1\}^n \to \{-1,1\}$ that has exactly 2 non-zero Fourier coefficients. By Parseval's theorem,

$$\widehat{f}(S_1)^2 + \widehat{f}(S_2)^2 = 1$$

Consider input $x \in \{0,1\}^n$ given by indices $S_1 \cup S_2$ set to 1 and all else set to -1 (we denote this as $x_{S_1 \cup S_2}$). Then,

$$\begin{aligned}
|f(x)| &= |\widehat{f}(S_1)\chi_{S_1}(x_{S_1\cup S_2}) + \widehat{f}(S_2)\chi_{S_2}(x_{S_1\cup S_2})| \\
&= |\widehat{f}(S_1) + \widehat{f}(S_2)| = 1
\end{aligned}$$

Similarly, there exist inputs such that $f(x) = \pm\widehat{f}(S_1) \pm \widehat{f}(S_2)$ all four combinations. Thus, any Fourier coefficients must in essence satisfy $\|f\|_1 = \|f\|_2$. The only solutions to this system are $(\pm 1, 0)$ and $(0, \pm 1)$. Thus, one of $\widehat{f}(S_1)$ or $\widehat{f}(S_2)$ is 0 and we have reached a contradiction.

(b) Suppose, for the sake of contradiction, that there exists an $f : \{-1,1\}^n \to \{-1,1\}$ that has exactly 3 non-zero Fourier coefficients. By Parseval's theorem,

$$\widehat{f}(S_1)^2 + \widehat{f}(S_2)^2 + \widehat{f}(S_3)^2 = 1$$

Consider input $x_{\cup_{i=1}^3 S_i}$

$$|f(x)| = |\sum_{i=1}^3 \widehat{f}(S_i)\chi_{S_i}(x_{\cup_{i=1}^3 S_i})| = |\sum_{i=1}^3 \widehat{f}(S_i)| = 1$$

Without loss of generality, assume $|S_1| \leq |S_2| \leq |S_3|$. We now claim that there exists an input x^* such that $\chi_{S_1}(x^*) = -\chi_{S_2}(x^*) = -\chi_{S_3}(x^*) = 1$. To see this, note that we can set $x_i^* = 1$ for all $i \in S_1$. Then, consider the following three cases

(1) $S_3 \backslash (S_1 \cup S_2) \neq \phi$ – $|S_2| \geq |S_1| \implies S_2 \backslash S_1 \neq \phi$. Thus, we can trivially make $\chi_{S_2}(x^*)$ odd parity by choosing $i \in S_2 \backslash S_1$ and setting $x_i^* = -1$. Similarly, since $S_3 \backslash (S_1 \cup S_2) \neq \phi$ we can appropriately set $j \in S_3 \backslash (S_1 \cup S_2)$ so as to give χ_{S_3} odd parity.

(2) $S_3 = (S_1 \cup S_2)$ – We set S_2 exactly as in case (1). Then, $\chi_{S_3}(x^*) = \chi_{S_1}(x^*) \cdot \chi_{S_2}(x^*) = -1$.

(3) $S_3 \subset (S_1 \cup S_2)$ – $|S_3| \geq |S_1| \implies S_3 \backslash S_1 \neq \phi \implies S_3 \cap S_2 \neq \phi$. Choose some $i \in S_3 \cap S_2$ and set $x_i^* = -1$ and set the remaining indices to 1. Then, $\chi_{S_2}(x^*) = \chi_{S_3}(x^*) = -1$.

Then,

$$|f(x^*)| = |\widehat{f}(S_1) - \widehat{f}(S_2) - \widehat{f}(S_3)| = 1$$

From above, however,

$$= |2\widehat{f}(S_1) - [\widehat{f}(S_1) + \widehat{f}(S_2) + \widehat{f}(S_3)]| = |2\widehat{f}(S_1) - 1| = 1$$

where we have assumed without loss of generality that $\widehat{f}(S_1) + \widehat{f}(S_2) + \widehat{f}(S_3) = 1$. $\widehat{f}(S_1) = 0$ and $\widehat{f}(S_1) = -1$ are the only two solutions. If $\widehat{f}(S_1) = 0$, f no longer has three non-zero coefficients and we have reached a contradiction. If $\widehat{f}(S_1) = -1$, then $\widehat{f}(S_2) = \widehat{f}(S_3) = 0$ by Parseval's theorem and we have reached another contradiction.

□

Problem 1.24 We can compute $||\varphi||_2^2$ directly:

$$\begin{aligned}
||\varphi||_2^2 &= \mathbb{E}_x[|\varphi(x)|^2] \\
&= \mathbb{E}_x[\varphi(x)^2] && \text{(nonnegativity of density)} \\
&= \frac{1}{2^n}\sum_{x\in\{-1,1\}^n} \varphi(x)^2 \\
&= \frac{1}{2^n}\sum_{x\in A} \varphi(x)^2 \\
&\geq \frac{1}{2^n}\sum_{x\in A}\left(\frac{2^n}{2^n\delta}\right)^2 && \text{(convexity of } f(z) = z^2\text{)} \\
&= \frac{1}{2^n}\frac{\delta 2^n}{\delta^2} \\
&= \frac{1}{\delta}.
\end{aligned}$$

□

Problem 1.27 (Necessary) Suppose, for the sake of contradiction that one could determine which linear function f is using only k queries, for $k < n$. However, there are at least 2^{n-k} linear functions that have the same mapping at these k inputs. There is no way to distinguish between these candidates given this information, so we have reached a contradiction and can conclude that n queries are necessary.

(Sufficient) Consider querying f on $\mathbf{1_k}$ where $\mathbf{1_k} = (1, 1, 1, ..., 0, ..., 1, 1)$ where the kth index is set to 0. Notice that $k \in S \iff f(\mathbf{1_k}) = 0$ and that there is $2^{n-n} = 1$ unique f with the given answers to these queries. □

Problem 1.28

(a) We have $\widehat{f}(S) \leq 2\delta \iff \text{dist}(f, \chi_S) \geq \frac{1}{2} - \delta$ and $\widehat{f}(S) \geq -2\delta \iff \text{dist}(f, \chi_S) \leq \frac{1}{2} + \delta$ in general. Now, take $S \neq S^*$. By the triangle inequality (equivalently a union bound), we have:

$$\text{dist}(f, \chi_S) \leq \text{dist}(\chi_S, \chi_{S^*}) + \text{dist}(f, \chi_{S^*}) = \frac{1}{2} + \delta$$

And:

$$\text{dist}(\chi_S, f) \geq \text{dist}(\chi_S, \chi_{S^*}) - \text{dist}(f, \chi_{S^*}) = \frac{1}{2} - \delta.$$

We conclude that $-2\delta \leq \widehat{f}(S) \leq 2\delta$ so $|\widehat{f}(S)| \leq 2\delta$, which is what we wanted to show.

(b) As in the chapter, we have $\Pr[\text{BLR accepts } f] = \frac{1}{2} + \frac{1}{2}\sum_{S\subseteq[n]} \widehat{f}(S)^3$. But now we have tighter control on $\widehat{f}(S)$ for all $S \neq S^*$. In particular, we can write:

$$\begin{aligned}
\frac{1}{2} + \frac{1}{2}\sum_{S\subseteq[n]} \widehat{f}(S)^3 &= \frac{1}{2} + \frac{1}{2}\widehat{f}(S^*)^3 + \frac{1}{2}\sum_{S\neq S^*} \widehat{f}(S)^3 \\
&\leq \frac{1}{2} + \frac{1}{2}(1-2\delta)^3 + \frac{1}{2}\max_{S\neq S^*}\{\widehat{f}(S)\} \cdot \sum_{S\neq S^*} \widehat{f}(S)^2 \\
&\leq \frac{1}{2} + \frac{1}{2}(1-2\delta)^3 + \delta(1-(1-2\delta)^2) \quad (\text{as } \widehat{f}(S) \leq |\widehat{f}(S)| \leq 2\delta \text{ for } S \neq S^*) \\
&= 1 - 3\delta + 10\delta^2 - 8\delta^3.
\end{aligned}$$

Therefore, $\Pr[\text{BLR rejects } f] \geq 3\delta - 10\delta^2 + 8\delta^3$ which is what we wanted to show. On a practical note, this implies that if the BLR Test rejects some function f with probability at most ε, then $\min_S \text{dist}(f, \chi_S) = \delta \leq \varepsilon/3$ approximately (ignoring higher-order terms). In other words, f is essentially $\varepsilon/3$-close to some linear function (stronger than the ε-closeness shown in the chapter).

(c) need to figure out

□

Problem 1.29

(a) $(\Leftarrow)$ Suppose $f(x) = a \cdot x$. Then,

$$f(x) + f(y) + f(z) = a \cdot x + a \cdot y + a \cdot z = a \cdot (x + y + z) = f(x + y + z)$$

$(\Rightarrow)$ Suppose $f(x) + f(y) + f(z) = f(x + y + z)$ and define $g(x) = f(x) + f(0)$. Then,

$$g(x + y) = f(x + y + 0) + f(0) = f(x) + f(y) + f(0) + f(0) = g(x) + g(y)$$

By the above, $g(x) = a \cdot x$. Thus, $f(x) = g(x) + f(0) = a \cdot x + f(0)$.

(b)

$$\begin{aligned}
\mathbb{E}_{x,y,z}[f(x)f(y)f(z)f(x+y+z)] &= \mathbb{E}_{x,y}[f(x)f(y)\mathbb{E}_z[f(z)f(x+y+z)]] \\
&= \mathbb{E}_{x,y}[f(x)f(y)(f * f)(x+y)] \quad \text{(Convolution definition)} \\
&= \mathbb{E}_x[f(x)\mathbb{E}_y[f(y)(f * f)(x+y)]] \\
&= \mathbb{E}_x[f(x)(f * f * f)(x)] \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)\widehat{f * f * f}(S) \quad \text{(Plancherel)} \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)\widehat{f * (f * f)}(S) \quad \text{(Associativity of Convolution)} \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)\widehat{f}(S)\widehat{f * f}(S) \quad \text{(Theorem 1.27)} \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)\widehat{f}(S)\widehat{f}(S)\widehat{f}(S) \quad \text{(Theorem 1.27)} \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)^4 \quad \text{(Theorem 1.27)}
\end{aligned}$$

(c) **Affine Test.** *Given query access to* $\mathbb{F}_2^n \to \mathbb{F}_2$:

- *Choose* $x, y, z \sim \mathbb{F}_2^n$ *independently*
- *Query* f *at* x, y, z, *and* $x + y + z$
- *"Accept" if* $f(x + y + z) = f(x) + f(y) + f(z)$

Claim. *Suppose Affine Test accepts* $f : \mathbb{F}_2^n \to \mathbb{F}_2$ *with probability* $1 - \epsilon$. *Then* f *is* ϵ*-close to being linear.*

We use the indicator function

$$\frac{1}{2} + \frac{1}{2}f(x)f(y)f(z)f(x+y+z) = \begin{cases} 1 \text{ if } f(x)f(y)f(z) = f(x+y+z) \\ 0 \text{ if } f(x)f(y)f(z) \neq f(x+y+z) \end{cases}$$

where the output of f is encoded as ± 1. Then,

$$\begin{aligned}
1 - \epsilon = \mathbb{P}_{x,y,z}[\textbf{Affine} \text{ accepts } f] &= \mathbb{E}_{x,y,z}[\frac{1}{2} + \frac{1}{2}f(x)f(y)f(z)f(x+y+z)] \\
&= \frac{1}{2} + \frac{1}{2}\sum_{S\subseteq[n]} \widehat{f}(S)^4 \quad \text{(part (b))}
\end{aligned}$$

Rearranging, we get

$$\begin{aligned}
1 - 2\epsilon &= \sum_{S\subseteq[n]} \widehat{f}(S)^4 \\
&\leq \left(\sum_{S\subseteq[n]} \widehat{f}(S)^2\right)\left(\sum_{S\subseteq[n]} \widehat{f}(S)^2\right) \\
&= \sum_{S\subseteq[n]} \widehat{f}(S)^2 \quad \text{(Parseval)} \\
&\leq \max_{S\subseteq[n]}\{\widehat{f}(S)\} \cdot \sum_{S\subseteq[n]} \widehat{f}(S) = \max_{S\subseteq[n]}\{\widehat{f}(S)\} \cdot f(\mathbf{1}) \\
&\leq \max_{S\subseteq[n]}\{\widehat{f}(S)\}
\end{aligned}$$

Notice that affine functions are of the form $\chi_S + b$ where χ_S is a linear function and $b \in \mathbb{F}_2^n$. Thus,

$$\begin{aligned}
\langle f, \chi_S + b\rangle &= \langle \sum_{T\subseteq[n]} \widehat{f}(T)\chi_T, \chi_S + b\rangle \\
&= \sum_{T\subseteq[n]} \widehat{f}(T)\langle \chi_T, \chi_S + b\rangle \\
&= \sum_{T\subseteq[n]} \widehat{f}(T)\langle \chi_T, \chi_S\rangle + \sum_{T\subseteq[n]} \widehat{f}(T)\langle \chi_T, b\rangle \\
&= \sum_{T\subseteq[n]} \widehat{f}(T)\langle \chi_T, \chi_S\rangle \\
&= \langle f, \chi_S\rangle = \widehat{f}(S)
\end{aligned}$$

Therefore, $1 - 2\text{dist}(f, \chi_S + b) = 1 - 2\text{dist}(f, \chi_S)$ meaning that $\text{dist}(f, \chi_S + b) = \text{dist}(f, \chi_S)$. Denote $S^* = \text{argmax}_{S\subseteq[n]}\{\widehat{f}(S)\}$. Then,

$$1 - 2\epsilon \leq 1 - \text{dist}(f, \chi_{S^*}) = 1 - 2\text{dist}(f, \chi_{S^*} + b)$$

which means that $\text{dist}(f, \chi_{S^*} + b) \leq \epsilon$ and thus that f is ϵ-close to the space of affine functions.

(d) $(\Rightarrow)$ Suppose f is affine. Then,

$$f(x + y) = f(x + y + 0) = f(x) + f(y) + f(0)$$

$(\Leftarrow)$ Suppose $f(x + y) = f(x) + f(y) + f(0)$. Then,

$$\begin{aligned}
f(x + y + z) &= f(x) + f(y + z) + f(0) \\
&= f(x) + f(y) + f(z) + f(0) + f(0) \\
&= f(x) + f(y) + f(z)
\end{aligned}$$

Then, the following 3-random affine test gives the same guarantees as the affine test and BLR test from 1.29(c) and the chapter.

3-Random Affine Test. *Given query access to* $\mathbb{F}_2^n \to \mathbb{F}_2$:

- *Choose* $x, y \sim \mathbb{F}_2^n$ *independently*
- *Query* f *at* x, y, *and* $x + y$
- *"Accept" if* $f(x + y) = f(x) + f(y) + f(0)$

□

2 Basic Concepts and Social Choice

Problem 2.2 Notice that the functions can be written as follows

(i) Majority: $\text{sgn}(x_1 + ... + x_n)$

(ii) AND: $\text{sgn}((n - \frac{1}{2}) + x_1 + ... + x_n)$

(iii) OR: $\text{sgn}((\frac{1}{2} - n) + x_1 + ... + x_n)$

(iv) $\pm\chi_i$: $\text{sgn}(\pm x_i)$

(v) ± 1: $\text{sgn}(\pm 1)$

□

Problem 2.3

(a) $(\Leftarrow)$ Suppose f is a weighted majority function so that we can write $f(x) = \text{sgn}(a_0 + x_1 + \ldots + x_n)$. f is clearly symmetric and monotone.

$(\Rightarrow)$ Suppose $f\colon \{-1, 1\}^n \to \{-1, 1\}$ is symmetric and monotone. By symmetry of f, we can write $f(x) = g(t(x))$ where $t(x)$ denotes the number of ones in x. Since f is monotone, we can write:

$$f(-1, \ldots, -1) \leq f(1, -1, \ldots, -1) \leq \cdots \leq f(1, \ldots, 1)$$

And thus

$$g(0) \leq \ldots \leq g(n).$$

It follows that $g(k) = \mathrm{sgn}(a+k)$ for some $a \in \mathbb{R}$. But for any x we have $x_1 + \ldots + x_n = t(x) - (n - t(x)) = 2t(x) - n$ and therefore $t(x) = \frac{1}{2}(n + x_1 + \ldots + x_n)$. Hence:

$$f(x) = g(t(x)) = \mathrm{sgn}\left(a + \frac{1}{2}(n + x_1 + \ldots + x_n)\right) = \mathrm{sgn}(2a + n + x_1 + \ldots + x_n)$$

Therefore f is a weighted majority function.

(b) Suppose $f\colon \{-1,1\}^n \to \{-1,1\}$ is symmetric, monotone and odd. By part (a), we can write $f(x) = \mathrm{sgn}(a_0 + x_1 + \ldots + x_n)$ for some $a_0 \in \mathbb{R}$. Since f is also odd, we have $\mathrm{sgn}(a_0 + x_1 + \ldots + x_n) = -\mathrm{sgn}(a_0 - x_1 - \ldots - x_n)$, implying that $|x_1 + \ldots + x_n| > |a_0|$ for all $x \in \{-1,1\}^n$. Note that the minimum value of $|x_1 + \ldots + x_n|$ is 0 is n is even and 1 is n is odd. Since $0 > |a_0|$ cannot hold, we must have n odd, in which case $1 > |a_0|$ implies that we can set $a_0 = 0$ without loss of generality. Hence $f = \mathrm{sgn}(x_1 + \ldots + x_n) = \mathrm{Maj}_n$.

$\square$

Problem 2.4 ($\Rightarrow$) Given a string $z \in \{-1,1\}^n$ and real r

$$\mathbf{1}_A = \mathrm{sgn}((n - 2r + 1) + \sum_i a_i x_i)$$

where $a_i = -z_i$. To see that this is true, notice that $\Delta(x,z) < r \iff \sum_i x_i z_i > (n-r+1)-(r-1) = n - 2r + 2$. Equivalently, $\Delta(x,z) < r \iff \sum_i x_i(-z_i) < -(n-r+1)+(r-1) = -(n-2r+2)$.

($\Leftarrow$) Given a linear threshold function of the form $f(x) = \mathrm{sgn}(a_0 + \sum_i a_i x_i)$ where $|a_1| = \ldots = |a_n|$, we show that it is an indicator function for some z and r. By the above, $z_i = -\frac{a_i}{|a_i|}$ and $r = \lceil \frac{n-a_0}{2} \rceil$. $\square$

Problem 2.11 For each coordinate $i \in S$

$$\mathbf{Inf}_i(f) = \sum_{T \ni i} \widehat{f}(T)^2 \geq \widehat{f}(S)^2 > 0$$

since $\widehat{f(S)} > 0$ by assumption. Therefore, all $i \in S$ have non-zero influence and are thus relevant. $\square$

Problem 2.12 Both computations are self-explanatory:

$$\begin{aligned}
\mathbb{E}_f[\mathbf{Inf}_i[f]] &= \mathbb{E}_f\left[\Pr_x[f(x) \neq f(x^{\oplus i})]\right] \\
&= \mathbb{E}_f\left[\mathbb{E}_x[\mathbf{1}_{\{f(x)\neq f(x^{\oplus i})\}}]\right] \\
&= \mathbb{E}_x\left[\mathbb{E}_f[\mathbf{1}_{\{f(x)\neq f(x^{\oplus i})\}}]\right] \\
&= \mathbb{E}_x[1/2] \\
&= \frac{1}{2}.
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}_f[\mathbf{I}[f]] &= \mathbb{E}_f\left[\sum_{i=1}^n \mathbf{Inf}_i[f]\right] \\
&= \sum_{i=1}^n \mathbb{E}_f[\mathbf{Inf}_i[f]] \\
&= \frac{n}{2}.
\end{aligned}$$

$\square$

Problem 2.13

(a) We compute the expected value and variance of f as follows

$$\begin{aligned}
\mathbb{E}[f] &= \mathbb{P}_x[f(x) = 1] - \mathbb{P}_x[f(x) = -1] \\
&= \left((1 - \frac{1}{2^w})^{2^w}\right) - \left(1 - (1 - \frac{1}{2^w})^{2^w}\right) \\
\lim_{w\to\infty} \mathbb{E}[f] &= \frac{1}{e} - (1 - \frac{1}{e}) = \frac{2}{e} - 1
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{Var}[f] &= \mathbb{E}[f^2] - \mathbb{E}[f]^2 \\
&= 1 - \mathbb{E}[f]^2 = 1 - \left(2\left((1 - \frac{1}{2^w})^{2^w}\right) - 1\right)^2 \\
&= 1 - \left(4\left((1 - \frac{1}{2^w})^{2^w}\right)^2 - 4\left((1 - \frac{1}{2^w})^{2^w}\right) + 1\right) \\
\lim_{w\to\infty} \mathbf{Var}[f] &= 1 - (\frac{4}{e^2} - \frac{4}{e} + 1) = \frac{4(e-1)}{e^2}
\end{aligned}$$

(b) Observe that the derivative

$$D_1 f = \frac{f^{(i\to 1)} - f^{(i\to -1)}}{2}$$

can never take value -1 for $f = \mathrm{Tribes}_{w,2^w}$ since the OR and AND functions are monotone (and the composition of monotone functions is monotone). This implies the following casework

$$D_1 f = \begin{cases} 0 \text{ if } \exists j \in [2, ..., 2^w].\ AND(x^j) = -1 \\ 0 \text{ if } \forall j \neq 1.\ AND(x^j) = 1 \text{ and } \exists k \in [2, ..., w] x^{1,k} = 1 \\ 1 \text{ otherwise} \end{cases}$$

(c) From part (b), $D_1 f = 1$ if $\forall j \neq 1.\ AND(x^j) = 1$ and $x^{1,k} = -1$ for $k = 2, ..., w$. Therefore,

$$\begin{aligned}
\mathbf{Inf}_1[f] &= \frac{2 \cdot (2^w - 1)^{2^w - 1}}{2^{w2^w}} \\
&= \frac{2 \cdot (2^w - 1)^{2^w}}{(2^w - 1)2^{w2^w}} \\
&= \left(\frac{2^w - 1}{2^w}\right)^{2^w} \cdot \frac{2}{2^w - 1} \\
\lim_{w\to\infty} \mathbf{Inf}_1[f] &= \frac{2}{e} \cdot \lim_{w\to\infty} \frac{1}{2^w - 1} = 0
\end{aligned}$$

Since each voter is equal, the total influence is given by

$$\begin{aligned}
\mathbf{I}[f] &= w2^w \mathbf{Inf}_i[f] = \left(\frac{2^w - 1}{2^w}\right)^{2^w} \cdot \frac{2w2^w}{2^w - 1} \\
\lim_{w\to\infty} \mathbf{I}[f] &= \infty
\end{aligned}$$

Interestingly, the influence of any given voter converges to 0 while the total influence diverges in the limit.

$\square$

Problem 2.20 Recall that the Laplacian operator L has the property $\mathrm{L}f = f(x) \cdot \mathbf{sens}_f(x)$ for $f\colon \{-1,1\}^n \to \{-1,1\}$. Therefore, we have:

$$\begin{aligned}
\mathbb{E}_x[\mathbf{sens}_f(x)^2] &= \mathbb{E}_x[\mathbf{sens}_f(x)f(x) \cdot \mathbf{sens}_f(x)f(x)] && (f^2 = 1) \\
&= \langle Lf, Lf \rangle \\
&= \langle \sum_{S\subseteq[n]} |S|\widehat{f}(S)\chi_S, \sum_{T\subseteq[n]} |T|\widehat{f}(T)\chi_T \rangle \\
&= \sum_{S,T\subseteq[n]} |S||T|\widehat{f}(S)\widehat{f}(T)\langle \chi_S, \chi_T \rangle \\
&= \sum_{S\subseteq[n]} |S|^2 \widehat{f}(S)^2 \\
&= \mathbb{E}_{S\sim\mathcal{S}_f}[|S|^2].
\end{aligned}$$

We remark that $\mathbb{E}_x[\mathbf{sens}_f(x)^3] \neq \mathbb{E}_{S\sim\mathcal{S}_f}[|S|^3]$ in general. For instance, take $f = \mathrm{Maj}_3$ and observe that $\mathbb{E}_x[\mathbf{sens}_f(x)^3] = \frac{1}{4} \cdot 0 + \frac{3}{4} \cdot 2^3 = 6$ whereas $\mathbb{E}_{S\sim\mathcal{S}_f}[|S|^3] = \frac{1}{4} \cdot (1 + 1 + 1 + 27) = \frac{15}{2}$. $\square$

Problem 2.22

(a)

$$\begin{aligned}
\mathbf{Inf}_i[\mathrm{Maj}_n] &= \Pr_x[\mathrm{Maj}_n(x) \neq \mathrm{Maj}_n(x^{\oplus i})] \\
&= \Pr_x\left[\sum_{j\neq i} x_j = 0\right] && (x_i \text{ must break tie}) \\
&= \binom{n-1}{(n-1)/2} \cdot 2^{1-n}.
\end{aligned}$$

(b) Let $n \in \mathbb{N}$ be odd. Then:

$$\frac{\mathbf{Inf}_1[\mathrm{Maj}_n]}{\mathbf{Inf}_1[\mathrm{Maj}_{n+2}]} = 4 \cdot \frac{\binom{n-1}{(n-1)/2}}{\binom{n+1}{(n+1)/2}} < 4 \cdot \frac{\binom{n-1}{(n-1)/2}}{4 \cdot \binom{n-1}{(n-1)/2}} = 1.$$

Here, the last inequality follows from:

$$\binom{n+1}{(n+1)/2} = \binom{n}{(n+1)/2} + \binom{n}{(n-1)/2} = \binom{n-1}{(n+1)/2} + 2 \cdot \binom{n-1}{(n-1)/2} + \binom{n-1}{(n-3)/2} < 4 \cdot \binom{n-1}{(n-1)/2}$$

due to maximality of central binomial coefficients. It follows that $\mathbf{Inf}_1[\mathrm{Maj}_n]$ is (strictly) decreasing for odd n.

(c)

$$\begin{aligned}
\mathbf{Inf}_1[\mathrm{Maj}_n] &= \binom{n-1}{(n-1)/2} \cdot 2^{1-n} \\
&= \frac{(n-1)!}{(((n-1)/2)!)^2} \cdot 2^{1-n} \\
&\approx \frac{n!}{((n/2)!)^2} \cdot 2^{-n} && \text{(asymptotically the same)} \\
&= \frac{(n/e)^n \cdot (\sqrt{2\pi n} + O(n^{-1/2}))}{(n/2e)^n \cdot (\sqrt{\pi n} + O((n/2)^{-1/2}))^2} \cdot 2^{-n} \\
&= \frac{\sqrt{2\pi n} + O(n^{-1/2})}{\pi n + O(1/n) + O(1)} \\
&= \frac{\sqrt{2\pi n} + O(n^{-1/2})}{\pi n} \\
&= \frac{\sqrt{2/\pi}}{\sqrt{n}} + O(n^{-3/2})
\end{aligned}$$

(d) Since Maj_n is monotone, we have $\mathbf{Inf}_i[\mathrm{Maj}_n] = \widehat{f}(i)$. Thus:

$$\begin{aligned}
\mathbf{W}^1[\mathrm{Maj}_n] &= \sum_{i=1}^{n} \widehat{f}(i)^2 \\
&= \sum_{i=1}^{n} \mathbf{Inf}_i[\mathrm{Maj}_n]^2 \\
&= n \cdot \left(\frac{\sqrt{2/\pi}}{\sqrt{n}} + O(n^{-3/2}) \right)^2 \\
&= n \cdot \left(\frac{2/\pi}{n} + O(n^{-3}) + \frac{4}{\pi} \cdot O(n^{-2}) \right) \\
&= \frac{2}{\pi} + O(n^{-2}) + \frac{4}{\pi} \cdot O(n^{-1}) \\
&\in \left[\frac{2}{\pi}, \frac{2}{\pi} + \frac{5}{\pi} \cdot n^{-1} \right] && \text{(for large enough } n\text{)}
\end{aligned}$$

Hence $2/\pi \leq \mathbf{W}^1[\mathrm{Maj}_n] \leq 2/\pi + O(n^{-1})$, as desired.

(e) Since Maj_n is symmetric, we have $\mathbf{W}^1[\mathrm{Maj}_n] = n \cdot \widehat{f}(1)^2$. By part (d), this implies

$$\widehat{f}(1)^2 \in \left[\frac{2}{\pi n}, \frac{2}{\pi n} + O(n^{-2}) \right].$$

Since $(\sqrt{2/\pi n} + O(n^{-3/2}))^2 = \frac{2}{\pi n} + O(n^{-2})) + O(n^{-3}) \geq \frac{2}{\pi n} + O(n^{-2}))$, we have:

$$\widehat{f}(1) \in \left[\sqrt{\frac{2}{\pi n}}, \sqrt{\frac{2}{\pi n}} + O(n^{-3/2}) \right].$$

Finally, we have:

$$\begin{aligned}
\mathbf{I}[\mathrm{Maj}_n] &= n \cdot \widehat{f}(1) \\
&\in \left[\sqrt{2/\pi} \cdot \sqrt{n}, \sqrt{2/\pi} \cdot \sqrt{n} + O(n^{-1/2}) \right].
\end{aligned}$$

□

Problem 2.23 Notice that if f is monotone, then by proposition 2.31

$$\mathbf{I}[f] = \sum_i \widehat{f(i)}$$

By the Cauchy-Schwarz inequality

$$\begin{aligned}
(\sum_i \widehat{f(i)} \cdot 1)^2 &\leq (\sum_i \widehat{f(i)}^2)(\sum_i 1^2) \\
&\leq n && \text{(Parseval's theorem)}
\end{aligned}$$

Therefore,

$$\sum_i \widehat{f(i)} \leq \sqrt{n}$$

for monotone f. □

Problem 2.27 Let x_1, x_2, x_3 be the three inputs that evaluate to 1 under f. If every dimension i-edge is a boundary edge for x_1, x_2, x_3, then the influence of coordinate i is maximized – namely $\mathbf{Inf}_i[f] = \frac{6}{2^n}$. For every coordinate to attain its maximum influence, it must be true that the pairwise hamming distances between x_1, x_2, x_3 are greater than 1 (if not, then there exists some coordinate i for which $x_1^{\oplus i} = x_2$ but $f(x_1) = f(x_2)$, meaning this dimension i-edge is not a boundary edge). Therefore,

$$\max_f \mathbf{I}[f] = \frac{6n}{2^n}$$

which is attained by f such that the pairwise hamming distances between x_1, x_2, x_3 are all greater than 1. □

Problem 2.28 Suppose f is even. Then $\widehat{f}(S) = 0$ whenever $|S|$ is odd. Thus $\mathbf{W}^1[f] = 0$ so:

$$\begin{aligned}
\mathbf{I}[f] &= \sum_{S \subseteq [n]} |S| \cdot \widehat{f}(S)^2 \\
&= \sum_{k=1}^{n} k \cdot \mathbf{W}^k[f] \\
&= \sum_{k=2}^{n} k \cdot \mathbf{W}^k[f] \\
&\geq 2 \sum_{k>0}^{n} \mathbf{W}^k[f] \\
&= 2 \cdot \mathbf{Var}[f].
\end{aligned}$$

Equivalently $\mathbf{Var}[f] \leq \frac{1}{2} \cdot \mathbf{I}[f]$, strengthening the Poincaré inequality. □

Problem 2.29

(a) Since $\mathbb{E}[f] = 0$, $f(\phi) = 0$. Note that

$$\mathbf{Var}[f] = \mathbb{E}[f^2] - \mathbb{E}[f]^2 = \mathbb{E}[f^2] = 1$$
$$\mathbf{Var}[f] = 1 = \sum_{k=1}^{n} \mathbf{W}^k[f] \leq \sum_{k=1}^{n} k\mathbf{W}^k[f] = \mathbf{I}[f] = \sum_{i=1}^{n} \mathbf{Inf}_i[f] \leq n \cdot \mathbf{MaxInf}[f]$$

Therefore,

$$1 \leq n\mathbf{MaxInf}[f]$$
$$\mathbf{MaxInf}[f] \geq \frac{1}{n}$$

(b) Observe that

$$\begin{aligned}
\mathbf{I}[f] &= \sum_{k=0}^{n} k\mathbf{W}^k[f] \geq \mathbf{W}^1[f] + 2(1 - \mathbf{W}^1[f]) \\
&= 2 - \mathbf{W}^1[f] \\
&= 2 - \sum_i \widehat{f(i)}^2 \\
&\geq 2 - \sum_i \mathbf{Inf}_i[f]^2 && \text{(Exercise 2.5)} \\
&\geq 2 - n\mathbf{MaxInf}[f]^2 && \text{(Monotonicity of the quadratic)}
\end{aligned}$$

Finally, we deduce that

$$\begin{aligned}
n\mathbf{MaxInf}[f] \geq \mathbf{I}[f] &\geq 2 - n\mathbf{MaxInf}[f]^2 \\
n\mathbf{MaxInf}[f]^2 + n\mathbf{MaxInf}[f] - 2 &\geq 0 \\
\mathbf{MaxInf}[f] &\geq \frac{-n + \sqrt{n^2 - 4(n)(-2)}}{2n} && \text{(Quadratic Formula)}
\end{aligned}$$

Thus,

$$\mathbf{MaxInf}[f] \geq \frac{n}{2} - \frac{1}{2} \geq \frac{2}{n} - \frac{4}{n^2}$$

□

Problem 2.31

(i) Since $f \geq 0$, $\mathbb{E}_{y \sim N_\rho(x)}[f(y)] \geq 0$

(ii) Notice that $f \geq 0, f \neq 0 \implies \exists z.\ f(z) > 0$. Since $p \in (-1, 1)$, $\mathbb{P}_{y \sim N_\rho(x)}[y = z] > 0$. Therefore, $\mathbb{E}_{y \sim N_\rho(x)}[f(y)] \geq \mathbb{P}_{y \sim N_\rho(x)}[y = z] \cdot f(z) > 0$.

□

Problem 2.32 Let $\rho_1, \rho_2 \in [-1, 1]$ and $f\colon \{-1, 1\} \to \mathbb{R}$. Observe that if $y \sim N_{\rho_1}(x)$ and $z \sim N_{\rho_2}(y)$, then for each $i \in [n]$ independently, we have:

$$\begin{aligned}
z_i &= \begin{cases} y_i & \text{with probability } \rho_2 \\ \text{unif. random} & \text{with probability } 1 - \rho_2 \end{cases} \\
&= \begin{cases} \begin{cases} x_i & \text{with probability } \rho_1\rho_2 \\ \text{unif. random} & \text{with probability } (1-\rho_1)\rho_2 \end{cases} \\ \\ \begin{cases} \text{unif. random} & \text{with probability } \rho_1(1-\rho_2) \\ \text{unif. random} & \text{with probability } (1-\rho_1)(1-\rho_2) \end{cases} \end{cases} \\
&= \begin{cases} x_i & \text{with probability } \rho_1\rho_2 \\ \text{unif. random} & \text{with probability } 1 - \rho_1\rho_2 \end{cases}
\end{aligned}$$

Hence $z \sim N_{\rho_1\rho_2}(x)$, which gives us:

$$\begin{aligned}
T_{\rho_1}T_{\rho_2}f(x) &= \mathbb{E}_{y \sim N_{\rho_1}(x)}[T_{\rho_2}f(y)] \\
&= \mathbb{E}_{y \sim N_{\rho_1}(x)}[\mathbb{E}_{z \sim N_{\rho_2}(y)}[f(z)]] \\
&= \mathbb{E}_{z \sim N_{\rho_1\rho_2}(x)}[f(z)] \\
&= T_{\rho_1\rho_2}f(x).
\end{aligned}$$

It follows that $T_{\rho_1}T_{\rho_2} = T_{\rho_1\rho_2}$ in general.

□

Problem 2.33 We show that the linear operator is a contraction on L^p by proving

$$\sum_x \mathbb{E}_{y \sim N_\rho(x)}[f(y)]^p \leq \sum_x f(x)^p$$

Observe that

$$\begin{aligned}
\sum_x \mathbb{E}_{y \sim N_\rho(x)}[f(y)]^p &= \sum_x \left(\sum_z f(z)\mathbb{P}_{y \sim N_\rho(x)}[y = z] \right)^p \\
&\leq \sum_x \sum_z f(z)^p\ \mathbb{P}_{y \sim N_\rho(x)}[y = z] && \text{(Jensen's inequality)} \\
&= \sum_z f(z)^p \sum_x \mathbb{P}_{y \sim N_\rho(x)}[y = z]
\end{aligned}$$

However,

$$\begin{aligned}
\sum_x \mathbb{P}_{y \sim N_\rho(x)}[y = z] &= \binom{n}{0}(\frac{1}{2} + \frac{1}{2}\rho)^n + \binom{n}{1}(\frac{1}{2} + \frac{1}{2}\rho)^{n-1}(\frac{1}{2} - \frac{1}{2}\rho)^1 + \ldots + \binom{n}{n}(\frac{1}{2} - \frac{1}{2}\rho)^n \\
&= \sum_{i=0}^{n} \binom{n}{i}(\frac{1}{2} + \frac{1}{2}\rho)^{n-i}(\frac{1}{2} - \frac{1}{2}\rho)^i \\
&= ((\frac{1}{2} + \frac{1}{2}\rho) + (\frac{1}{2} - \frac{1}{2}\rho))^n = 1^n = 1 && \text{(Binomial Theorem)}
\end{aligned}$$

Where we think of the quantity $\sum_x \mathbb{P}_{y\sim N_\rho(x)}[y=z]$ as the partitioned sum over x distance k from z. Then, for y ρ-correlated with x, there must be k bit flips and $n-k$ bit preservations. Then,

$$\begin{aligned}\sum_x \mathbb{E}_{y\sim N_\rho(x)}[f(y)]^p &\le \sum_z f(z)^p \sum_x \mathbb{P}_{y\sim N_\rho(x)}[y=z]\\ &= \sum_z f(z)^p\end{aligned}$$

and we can conclude that $\|T_\rho f\|_p \le \|f\|_p$. □

Problem 2.34 Let $f\colon \{-1,1\}\to\mathbb{R}$ and $\rho\in[-1,1]$. For any $x\in\{-1,1\}$ we have:

$$\begin{aligned}|T_\rho f(x)| &= \left|\mathbb{E}_{y\sim N_\rho(x)}[f(y)]\right|\\ &\le \mathbb{E}_{y\sim N_\rho(x)}[|f(y)|] && \text{(convexity of } |\cdot|)\\ &= T_\rho|f|(x).\end{aligned}$$

Suppose $\rho\in(-1,1)$ so that for any $z\in\{-1,1\}$ we have $\Pr_{y\sim N_\rho(x)}[y=z]>0$. We observe that equality above holds if and only if $\operatorname{sgn}(f(y))=\operatorname{sgn}(f(z))$ for all $y,z\in\{-1,1\}^n$ where $f(y),f(z)\neq 0$, meaning $f\ge 0$ everywhere or $f\le 0$ everywhere. Another way of seeing this is that equality in Jensen's holds only if $\varphi(t)=|t|$ is linear on the support of f, meaning the support of f is either contained in $\mathbb{R}^{\ge 0}$ or contained in $\mathbb{R}^{\le 0}$. □

Problem 2.41 Observe that

$$\frac{d^2}{d\rho^2}\mathbf{Stab}_\rho[f] = \sum_{k=0}^n k(k-1)\rho^{k-2}\mathbf{W}^k[f]\ge 0$$

for $\rho\in[0,1]$. □

Problem 2.42 Let $f\colon\{-1,1\}^n\to\{-1,1\}$ and let $\delta\in[0,1]$. Let $x\sim\{-1,1\}^n$ uniformly, and for each $i\in[n]$ independently let $y_i=\begin{cases}-x_i & \text{w.p. } \delta\\ x_i & \text{w.p. } 1-\delta\end{cases}$. Then $\mathbf{NS}_\delta[f] := \Pr_{x,y}[f(x)\neq f(y)]$. Now, define $y^{(0)}=(x_1,\dots,x_n)=x$ and for each $i\in[n]$ define $y^{(i)}=(y_1,\dots,y_i,x_{i+1},\dots,x_n)$ so that $y^{(n)}=y$. We make the simple observation that $f(x)\neq f(y)$ implies that $f(y^{(i)})\neq f(y^{(i-1)})$ for some $i\in[n]$. Moreover, since x is uniformly random, the $y^{(i)}$ are individually uniformly random as

well. By construction, we have $y^{(i)}=\begin{cases}(y^{(i-1)})^{\oplus i} & \text{w.p. } \delta\\ y^{(i-1)} & \text{w.p. } 1-\delta\end{cases}$. Therefore, we have:

$$\begin{aligned}\mathbf{NS}_\delta[f] &= \Pr[f(x)\neq f(y)]\\ &= \Pr\left[\bigcup_{i=1}^n \{f(y^{(i)})\neq f(y^{(i-1)})\}\right]\\ &\le \sum_{i=1}^n \Pr\Big[f(y^{(i-1)})\neq f(y^{(i)})\Big]\\ &= \sum_{i=1}^n \Pr\Big[f(y^{(i-1)})\neq f(y^{(i)}) \mid y^{(i-1)}\neq y^{(i)}\Big]\cdot\Pr\Big[y^{(i-1)}\neq y^{(i)}\Big]\\ &= \sum_{i=1}^n \Pr\Big[f(y^{(i-1)})\neq f(y^{(i)}) \mid y^{(i-1)}\neq y^{(i)}\Big]\cdot\delta\\ &= \delta\cdot\sum_{i=1}^n \Pr_{z\sim\{-1,1\}^n}[f(z)\neq f(z^{\oplus i})]\\ &= \delta\cdot\sum_{i=1}^n \mathbf{Inf}_i[f]\\ &= \delta\cdot\mathbf{I}[f].\end{aligned}$$

□

Problem 2.45 Let $0<\delta\le 1$ and $k\in\mathbb{N}^+$. Observe the following inequalities:

$$\sum_{j=0}^{k-1}(1-\delta)^j \ge \sum_{j=0}^{k-1}(1-\delta)^{k-1} = (1-\delta)^{k-1}k$$

And

$$\sum_{j=0}^{k-1}(1-\delta)^j \le \sum_{j=0}^{\infty}(1-\delta)^j = \frac{1}{1-(1-\delta)} = \frac{1}{\delta}.$$

Hence $(1-\delta)^{k-1}k\le\frac{1}{\delta}$. □

Problem 2.46 Let $f\colon\{-1,1\}^n\to\mathbb{R}$ and $0\le\rho-\varepsilon\le\rho<1$ For $t\in(0,1)$ let $g(t)=\dfrac{d}{dt}\mathbf{Stab}_t[f]=\sum_{k=1}^n kt^{k-1}\mathbf{W}^k[f]$. If $\varepsilon=0$ then the inequality is trivial. Otherwise, by the mean value theorem, we can find $c\in(\rho-\varepsilon,\rho)$ for which $|\mathbf{Stab}_\rho[f]-\mathbf{Stab}_{\rho-\varepsilon}[f]|=\varepsilon|g(c)|$. Thus, it suffices to show that

$|g(c)|\le\frac{1}{1-\rho}\cdot\mathbf{Var}[f]$, which we can do as follows:

$$\begin{aligned}|g(c)| &= \left|\frac{d}{dt}\Big|_{t=c}\mathbf{Stab}_t[f]\right|\\ &= \sum_{k=1}^n kc^{k-1}\mathbf{W}^k[f]\\ &= \left|\frac{d}{dt}\Big|_{t=c}\mathbf{Stab}_t[f]\right|\\ &= \sum_{k=1}^n \frac{1}{1-c}\cdot\mathbf{W}^k[f]\\ &= \left|\frac{d}{dt}\Big|_{t=c}\mathbf{Stab}_t[f]\right|\\ &= \frac{1}{1-c}\cdot\mathbf{Var}[f]\\ &\le \frac{1}{1-\rho}\cdot\mathbf{Var}[f]. && \text{(as } c\ge\rho)\end{aligned}$$

□

Problem 2.54 We start with the base case $n=1$. We have:

$$\mathbf{Var}[f]=\left(\frac{f(1)-f(-1)}{2}\right)^2=\mathbb{E}_x[\mathrm{D}_i f(x)^2]=\mathbf{I}[f].$$

Now, suppose assume the inequality holds for n, and let $f\colon\{-1,1\}^{n+1}\to\{-1,1\}$. Consider $g,h\colon\{-1,1\}^n\to\mathbb{R}$ given by $g(x_1,\dots,x_n)=f(x_1,\dots,x_n,-1)$ and $h(x_1,\dots,x_n)=f(x_1,\dots,x_n,1)$. Then, we have:

$$\mathbb{E}[f^2]=\frac{1}{2}\cdot\mathbb{E}[g^2+h^2]=\frac{1}{2}\cdot\mathbb{E}[g^2]+\frac{1}{2}\cdot\mathbb{E}[h^2]$$

And:

$$\mathbb{E}[f]^2=\left(\frac{1}{2}\cdot\mathbb{E}[g]+\frac{1}{2}\cdot\mathbb{E}[h]\right)^2=\frac{1}{4}\cdot\mathbb{E}[g]^2+\frac{1}{4}\cdot\mathbb{E}[h]^2+\frac{1}{2}\cdot\mathbb{E}[g]\cdot\mathbb{E}[h].$$

Moreover, we notice that for $1\le i\le n$ we have $\mathbf{Inf}_i[f]=\frac{1}{2}\cdot\mathbf{Inf}_i[g]+\frac{1}{2}\cdot\mathbf{Inf}_i[h]$, whereas $\mathbf{Inf}_{n+1}[f]=\mathbb{E}[z^2]$ where $z:=\mathrm{D}_{n+1}f=\frac{g-h}{2}$. Therefore, we can write:

$$\mathbf{I}[f]=\frac{1}{2}\cdot\mathbf{I}[g]+\frac{1}{2}\cdot\mathbf{I}[h]+\mathbb{E}[z^2]$$

We now prove the desired inequality:

$$\begin{aligned}\mathbf{Var}[f] &= \mathbb{E}[f^2]-\mathbb{E}[f]^2\\ &= \left(\frac{1}{2}\cdot\mathbb{E}[g^2]+\frac{1}{2}\cdot\mathbb{E}[h^2]\right)-\left(\frac{1}{4}\cdot\mathbb{E}[g]^2+\frac{1}{4}\cdot\mathbb{E}[h]^2+\frac{1}{2}\cdot\mathbb{E}[g]\cdot\mathbb{E}[h]\right)\\ &= \frac{1}{2}\cdot\left(\mathbb{E}[g^2]-\mathbb{E}[g]^2\right)+\frac{1}{2}\cdot\left(\mathbb{E}[h^2]-\mathbb{E}[h]^2\right)+\frac{1}{4}\cdot\left(\mathbb{E}[g]^2+\mathbb{E}[h]^2-2\cdot\mathbb{E}[g]\cdot\mathbb{E}[h]\right)\\ &= \frac{1}{2}\cdot\mathbf{Var}[g]+\frac{1}{2}\cdot\mathbf{Var}[h]+\mathbb{E}[z]^2 && (z=\tfrac{g-h}{2})\\ &\le \frac{1}{2}\cdot\mathbf{I}[g]+\frac{1}{2}\cdot\mathbf{I}[h]+\mathbb{E}[z]^2 && \text{(induction hypothesis)}\\ &= \mathbf{I}[f]-\mathbb{E}[z^2]+\mathbb{E}[z]^2\\ &\le \mathbf{I}[f] && \text{(Jensen's inequality)}\end{aligned}$$

By induction, we have proved the Poincaré inequality for all $n\in\mathbb{N}$. □

Problem 2.55

(a) By definition of the Laplacian operator

$$\begin{aligned}Lg(x) &= \sum_{i=1}^n L_i g(x)=\sum_{i=1}^n\frac{g(x)-g(x^{\oplus i})}{2}\\ &= \frac{n}{2}g(x)-\frac{1}{2}\sum_{i=1}^n \| -x_iw_i+\sum_{j\neq i}^n x_jw_j\|\\ &\le \frac{n}{2}g(x)-\frac{1}{2}\|\sum_{i=1}^n\left[-x_iw_i+\sum_{j\neq i}^n x_jw_j\right]\| && \text{(Triangle Inequality)}\\ &= \frac{n}{2}g(x)-\frac{1}{2}\|(n-2)\sum_{i=1}^n x_iw_i\|\\ &= \frac{n-(n-2)}{2}g(x)=g(x) && \text{(Property of } \|\cdot\| \text{ norm)}\end{aligned}$$

since $x\in\{-1,1\}^n$ was chosen arbitrarily, we can conclude that the Laplacian of g is pointwise smaller than g.

(b) We first note that g is an even function. This is because

$$g(-x)=\|\sum_{i=1}^n -x_iw_i\|=\|\sum_{i=1}^n x_iw_i\|=g(x) \qquad \text{(Property of } \|\cdot\| \text{ norm)}$$

Then, by exercise 2.28 we have

$$
\begin{aligned}
2\mathbf{Var}[g] &\leq \mathbf{I}[g] = \sum_{i=1}^{n} \mathbf{Inf}_i[g] \\
&= \sum_{i=1}^{n} \langle g, L_i g\rangle && \text{(Proposition 2.26)} \\
&= \langle g, \sum_{i=1}^{n} L_i g\rangle = \langle g, Lg\rangle && \text{(Bilinearity of } \langle\cdot,\cdot\rangle) \\
&\leq \langle g, g\rangle = \mathbb{E}[g^2] && \text{(Part (a))}
\end{aligned}
$$

We can now deduce the *Khintchine-Kahane* inequality

$$
\begin{aligned}
2\mathbf{Var}[g] = 2\mathbb{E}[g^2] - 2\mathbb{E}[g]^2 &\leq \mathbb{E}[g^2] \\
\mathbb{E}[g^2] &\leq 2\mathbb{E}[g]^2 \\
\mathbb{E}[g] &\geq \frac{1}{\sqrt{2}}\mathbb{E}[g^2]^{\frac{1}{2}}
\end{aligned}
$$

Expanding, we get

$$
\mathbb{E}_x[\|\sum_{i=1}^{n} x_i w_i\|] \geq \frac{1}{\sqrt{2}}\mathbb{E}_x[\|\sum_{i=1}^{n} x_i w_i\|^2]^{\frac{1}{2}}
$$

(c) To see that the bound is tight, we construct an example that achieves equality. Consider $V = \mathbb{R}$, $n = 2$, and $w_1 = w_2 = 1$. Here, we get that

$$
\begin{aligned}
\mathbb{E}_x[|x_1 + x_2|] &= \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot 0 = 1 \\
\mathbb{E}_x[|x_1 + x_2|^2]^{\frac{1}{2}} &= \sqrt{\frac{1}{2}\cdot 4 + \frac{1}{2}\cdot 0} = \sqrt{2} \\
\implies \mathbb{E}_x[|x_1 + x_2|] &= \frac{1}{\sqrt{2}}\mathbb{E}_x[|x_1 + x_2|^2]^{\frac{1}{2}}
\end{aligned}
$$

□

3 Spectral Structure and Learning

Problem 3.2

□

Problem 3.3

$$
\begin{aligned}
\mathbb{E}[f^2] - \mathbf{Stab}_{1-\delta}[f] &= \sum_S \left[1 - (1-\delta)^{|S|}\right]\widehat{f}(S)^2 \\
&= \sum_{k=1}^{} n\left[1 - (1-\delta)^k\right]\mathbf{W}^k[f] \\
&\geq \left(1 - (1-\delta)^{\frac{1}{\delta}}\right)\mathbf{W}^k[f] \\
&\geq (1 - \frac{1}{e})\Pr\left[|S| \geq \frac{1}{\delta}\right]
\end{aligned}
$$

Therefore,

$$
\Pr\left[|S| \geq \frac{1}{\delta}\right] \leq \frac{\mathbb{E}[f^2] - \mathbf{Stab}_{1-\delta}[f]}{1 - \frac{1}{e}}
$$

□

Problem 3.4 We prove the claim inductively. For, $n = k$ it is trivially true that $\mathbb{P}_x[f(x) \neq 0] \geq 2^{-n} = 2^{-k}$ since f is not identically 0. We assume that for $f : \{-1,1\}^n \to \mathbb{R}$ where $\deg(f) \leq k$ we have $\mathbb{P}_x[f(x) \neq 0] \geq 2^{-k}$. Then, consider $f : \{-1,1\}^{n+1} \to \mathbb{R}$ such that $\deg(f) \leq k$ and its corresponding sub-functions $f_1 = f(x_1, ..., x_n, 1)$ and $f_2 = f(x_1, ..., x_n, -1)$. There are then two cases

1. Sub-functions f_1 and f_2 are both not identically 0. Notice that because $\deg(f) \leq k$, it must be the case that $\deg(f_1), \deg(f_2) \leq k$. Then, by the inductive hypothesis,

$$
\begin{aligned}
\mathbb{P}_{x\in\{-1,1\}^n}[f_1(x) \neq 0] \geq 2^{-k} \\
\mathbb{P}_{x\in\{-1,1\}^n}[f_2(x) \neq 0] \geq 2^{-k}
\end{aligned}
$$

   Notice that

$$
\mathbb{P}_{x\in\{-1,1\}^{n+1}}[f(x) \neq 0] = \frac{1}{2}\left(\mathbb{P}_{x\in\{-1,1\}^n}[f_1(x) \neq 0] + \mathbb{P}_{x\in\{-1,1\}^n}[f_2(x) \neq 0]\right) \geq 2^{-k}
$$

2. Without loss of generality, sub-function f_2 is identically 0. By the definition of differentiation

$$
D_{n+1}f = \frac{f(x^{(n+1)\mapsto 1}) - f(x^{(n+1)\mapsto -1})}{2} = \frac{f_1 - f_2}{2} = \frac{f_1}{2}
$$

   By the differentiation formula in chapter 2, $\deg(g) \leq k \implies \deg(D_i g) \leq k - 1$. Therefore, $\deg(\frac{f_1}{2}) = \deg(f_1) = \deg(D_{n+1}f) \leq k - 1$. By the inductive hypothesis then, $\mathbb{P}_x[f_1(x) \neq 0] \geq$

2^{1-k}. We can then conclude that

$$
\begin{aligned}
\mathbb{P}_{x\in\{-1,1\}^{n+1}}[f(x) \neq 0] &= \frac{1}{2}\left(\mathbb{P}_{x\in\{-1,1\}^n}[f_1(x) \neq 0] + \mathbb{P}_{x\in\{-1,1\}^n}[f_2(x) \neq 0]\right) \\
&\geq \frac{1}{2}\cdot 2^{1-k} = 2^{-k}
\end{aligned}
$$

Note that both sub-functions cannot be identically 0 because this would contradict the assumption that f is not identically 0. The two above cases are then exhaustive and show the claim.

□

Problem 3.7

(a)

$$
\begin{aligned}
\hat{\lVert} f_{J|z} \hat{\rVert}_1 &= \sum_{S\subseteq J} |\widehat{f}_{J|z}(S)| \\
&= \sum_{S\subseteq J}\left|\sum_{T\subseteq\overline{J}} \widehat{f}(S\cup T)z^T\right| \\
&\leq \sum_{S\subseteq J}\sum_{T\subseteq\overline{J}} |\widehat{f}(S\cup T)z^T| \\
&= \sum_{S\subseteq J}\sum_{T\subseteq\overline{J}} |\widehat{f}(S\cup T)| \\
&= \sum_{S\subseteq[n]} |\widehat{f}(S)| && (S = (S\cap J)\cup(S\cap\overline{J})) \\
&= \hat{\lVert} f \hat{\rVert}_1
\end{aligned}
$$

(b) Let $\widehat{f}_{J|z}(S) \neq 0$. Then $\sum_{T\subseteq\overline{J}} \widehat{f}(S\cup T)z^T \neq 0$ meaning $\widehat{f}(S\cup T) \neq 0$ for some $T \subseteq \overline{J}$. Now, observe that if $S, S' \subseteq J$ and $T, T' \subseteq \overline{J}$ then $S\cup T = S'\cup T' \iff S = S', T = T'$. In particular, each unique S for which $\widehat{f}_{J|z}(S) \neq 0$ corresponds to at least one $R = S\cup T$ for which $\widehat{f}(R) \neq 0$, and the same R will not be found for multiple sets S. We conclude that $\text{sparsity}(\widehat{f_{J|z}}) \leq \text{sparsity}(\widehat{f})$, as desired.

□

Problem 3.10 Consider the Fourier expansion of $D_i f$:

$$
D_i f(x) = \sum_{\substack{S\subseteq[n] \\ S\ni i}} .
$$

Thus, we can write $\widehat{D_i f}(S) = \begin{cases} 0 & \text{if } i \in S \\ \widehat{f}(S\cup\{i\}) & \text{if } i \notin S \end{cases}$. From Exercise 3.9, we know that $\hat{\lVert} D_i f \hat{\rVert}_\infty \leq \|D_i f\|_1$, which gives us:

$$
\begin{aligned}
\max_{S\ni i} |\widehat{f}(S)| &\leq \mathbb{E}_x[D_i f(x)] \\
&= \mathbb{E}_x[D_i f(x)^2] && (D_i f \to \{0,1\} \text{ since } f \text{ monotone}) \\
&= \mathbf{Inf}_i[f] \\
&= \widehat{f}(i).
\end{aligned}
$$

It follows that $\hat{\lVert} f \hat{\rVert}_\infty$ is achieved by an S of cardinality 0 or 1. □

Problem 3.13

(a) A must obviously be nonempty, so let $a \in A$ be arbitrary. Since A is not affine, $A + a$ is not a subspace of $\mathbb{F}_2^n$, meaning there exist $b, c \in A + a$ for which $b + c \notin A + a$. Hence, $a + b, a + c \in A$ but $a + b + c \notin A$. Then if $B = \{0, b, c, b + c\} + a = \{a, a + b, a + c, a + b + c\}$, we have that B is an affine subspace of dimension two, and it intersects with A on three points (all except $a + b + c$).

(b) For the B obtained in (a), write $B = H + a$ for some subspace $H \leq \mathbb{F}_2^n$ and let $B\backslash A = \{b\}$. As shown previously, we have $\widehat{\varphi_B}(S) = \begin{cases} \chi_S(a) & \text{if } S \in H^\perp \\ 0 & \text{otherwise} \end{cases}$, and $\widehat{\varphi_b}(S) = \chi_S(a)$. Thus, $|\widehat{\psi}(S)| = |\widehat{\varphi_B}(S) - \frac{1}{2}\widehat{\varphi_b}(S)| = |\pm\frac{1}{2}\chi_S(a)| = 1/2$, meaning $\hat{\lVert}\psi\hat{\rVert}_\infty = \frac{1}{2}$.

(c) We have:

$$
\begin{aligned}
\langle\psi, f\rangle &= \langle\varphi_B, f\rangle - \frac{1}{2}\cdot\langle\varphi_b, f\rangle \\
&= \mathbb{E}_{x\sim B}[f(x)] - \frac{1}{2}\cdot f(b) \\
&= \mathbb{E}_{x\sim B}[f(x)] && (b \notin A) \\
&= 3/4.
\end{aligned}
$$

   Then, we conclude:

$$
\begin{aligned}
\hat{\lVert} f \hat{\rVert}_1 &= \sum_{S\subseteq[n]} |\widehat{f}(S)| \\
&\geq 2\cdot\sum_{S\subseteq[n]} \frac{1}{2}\cdot\widehat{f}(S) \\
&\geq 2\cdot\sum_{S\subseteq[n]} \widehat{\psi}(S)\cdot\widehat{f}(S) && \text{(part (b))} \\
&= 2\cdot\langle\psi, f\rangle && \text{(Plancherel)} \\
&= 3/2.
\end{aligned}
$$

□

Problem 3.14

$$
\begin{aligned}
\hat{\|} f \hat{\|}_1 = \sum_{S \subseteq [n]} |\widehat{f}(S)| &\leq \left(\sum_{S \subseteq [n]} \widehat{f}(S)^2 \right)^{\frac{1}{2}} \left(\sum_{S \subseteq [n]} 1^2 \right)^{\frac{1}{2}} && \text{(Cauchy Schwartz)} \\
&\leq 1 \cdot 2^{n/2} && (\mathbb{E}[f^2] \leq 1)
\end{aligned}
$$

Now, consider the inner product function $IP^{2n} : \{-1, 1\}^{2n} \to \{-1, 1\}$. From exercise 1.1, we know that $|\widehat{IP}(S)| = \frac{1}{2^n}$ for all $S \subseteq [2n]$. Therefore,

$$
\begin{aligned}
\hat{\|} IP \hat{\|}_1 &= \sum_{S \subseteq [2n]} |\widehat{IP}(S)| \\
&= \frac{1}{2^n} \cdot 2^{2n} = 2^n = 2^{2n/2}
\end{aligned}
$$

Therefore, for any $n \in \mathbb{N}$, IP^{2n} achieves equality (which also implies the bound is tight). □

Problem 3.16 Consider set $\mathcal{F} = \{S | \widehat{f}(S) \geq \frac{\epsilon}{\hat{\|} f \hat{\|}_1}\}$. We first show that $|\mathcal{F}| \leq \frac{\hat{\|} f \hat{\|}_1^2}{\epsilon}$. Suppose, for the sake of contradiction that $|\mathcal{F}| > \frac{\hat{\|} f \hat{\|}_1^2}{\epsilon}$. Then,

$$
\sum_{S \in \mathcal{F}} \widehat{f}(S)^2 \geq |\mathcal{F}| \cdot \min_{S \in \mathcal{F}} \widehat{f}(S)^2 > \frac{\hat{\|} f \hat{\|}_1^2}{\epsilon} \cdot \frac{\epsilon}{\hat{\|} f \hat{\|}_1} = \hat{\|} f \hat{\|}_1
$$

Contradiction. Finally, note that

$$
\begin{aligned}
\sum_{S \notin \mathcal{F}} \widehat{f}(S)^2 &\leq \max_{S \notin \mathcal{F}} \widehat{f}(S) \cdot \sum_{S \notin \mathcal{F}} \widehat{f}(S) \\
&\leq \frac{\epsilon}{\hat{\|} f \hat{\|}_1} \cdot \hat{\|} f \hat{\|}_1 = \epsilon
\end{aligned}
$$

and we can conclude that f is ϵ-concentrated on $\mathcal{F}$. □

Problem 3.17 For any S, we have $|\widehat{g}(S)| \leq |\widehat{g-f}(S)| + |\widehat{f}(S)|$. Squaring both sides, and summing over $\overline{\mathcal{F}}$, we have:

$$
\begin{aligned}
\sum_{S \notin \mathcal{F}} \widehat{g}(S)^2 &\leq \sum_{S \notin \mathcal{F}} (|\widehat{g-f}(S)| + |\widehat{f}(S)|)^2 \\
&\leq \sum_{S \notin \mathcal{F}} \widehat{g-f}(S)^2 + \sum_{S \notin \mathcal{F}} \widehat{f}(S)^2 + 2 \cdot \sum_{S \notin \mathcal{F}} |\widehat{g-f}(S)| \cdot |\widehat{f}(S)| \\
&\leq \hat{\|} g - f \hat{\|}_2^2 + \sum_{S \notin \mathcal{F}} \widehat{f}(S)^2 + 2 \cdot \sum_{S \notin \mathcal{F}} |\widehat{g-f}(S)| \cdot |\widehat{f}(S)| \\
&= \|g - f\|_2^2 + \sum_{S \notin \mathcal{F}} \widehat{f}(S)^2 + 2 \cdot \sum_{S \notin \mathcal{F}} |\widehat{g-f}(S)| \cdot |\widehat{f}(S)| && \text{(Parseval's)} \\
&\leq \varepsilon_1 + \varepsilon_2 + 2 \cdot \sum_{S \notin \mathcal{F}} |\widehat{g-f}(S)| \cdot |\widehat{f}(S)| \\
&\leq \varepsilon_1 + \varepsilon_2 + 2 \sqrt{\left(\sum_{S \notin \mathcal{F}} \widehat{g-f}(S)^2 \right) \cdot \left(\sum_{S \notin \mathcal{F}} \widehat{f}(S)^2 \right)} && \text{(Cauchy-Schwarz)} \\
&\leq \varepsilon_1 + \varepsilon_2 + 2\sqrt{\varepsilon_1 \varepsilon_2} && \text{(same bounds as before)} \\
&\leq 2(\varepsilon_1 + \varepsilon_2). && \text{(AM-GM inequality)}
\end{aligned}
$$

Hence, g is $2(\varepsilon_1 + \varepsilon_2)$-concentrated on $\mathcal{F}$, as desired. □

Problem 3.19 We assume that f can be computed by decision tree T. Then, notice that $-f(x)$ can be computed by $-T$, where $-T$ flips the outputs at leaf nodes of tree T. Decision tree $-T$ retains size s and depth k. Furthermore, the dual of f is defined as $f^\dagger = -f(-x)$. It suffices to show that $f(-x)$ can be computed by decision tree of size s and depth k. Decision tree $T(-)$ that flips every value along an edge of tree T computes $f(-x)$. Decision tree $T(-)$ retains size s and depth k. Therefore, decision tree $-T(-)$ computes $f^\dagger$ with size s and depth k. □

Problem 3.22 We observe that $T'(x) \neq T(x)$ only if $|C_p(x)| > \log(s/\varepsilon)$. Since a computation path of length ℓ is followed by at most $2^{n-\ell}$ inputs, each truncated computation path is followed by at most $2^{n - \log(s/\varepsilon)} = 2^n \cdot \frac{\varepsilon}{s}$ inputs. Since the number of computation paths is the number of leaves, namely s, it follows that $T'(x) \neq T(x)$ can only occur on at most $s \cdot 2^n \cdot \frac{\varepsilon}{s} = \varepsilon 2^n$ inputs, meaning the function computed by T' is ε-close to the function computed by T. □

Problem 3.23 Given function f that is computable by a decision tree, we construct a linear threshold function and prove its equivalence with f. By definition, a decision list only contains one path of internal nodes, which we denote $x_{i_0}, ..., x_{i_k}$ where x_{i_l} is at level l. Notice that $x_{i_0}, ..., x_{i_{k-1}}$ each connect to exactly one leaf while x_{i_k} connects to two leaves. To see this, suppose that x_{i_l} is the first internal node in the decision list that connects to two leaves for some $l \in \{0, ..., k-1\}$. Then, internal nodes $x_{i_{l+1}}$ onwards cannot exist. Conversely, if x_{i_k} did not connect to two leaf nodes, then another internal node $x_{i_{k+1}}$ must exist.

Now consider x_{i_l} and define t_l corresponding to leaf edges as follows

$$
t_l = \begin{cases}
\frac{x_{i_l}+1}{2} \cdot 2^{n-l} \text{ if } x_{i_l} = 1 \implies f(x) = 1 \\
\frac{x_{i_l}+1}{2} \cdot -2^{n-l} \text{ if } x_{i_l} = 1 \implies f(x) = -1 \\
\frac{x_{i_l}-1}{2} \cdot -2^{n-l} \text{ if } x_{i_l} = -1 \implies f(x) = 1 \\
\frac{x_{i_l}-1}{2} \cdot 2^{n-l} \text{ if } x_{i_l} = -1 \implies f(x) = -1
\end{cases}
$$

Then, let $g(x) = \text{sgn}(\sum_{l=0}^{k-1} t_l + t_k^{(1)} + t_k^{(2)})$ where $t_k^{(1)}$ and $t_k^{(2)}$ denote the two leaf edges for x_{i_k}. We claim that $\forall x. g(x) = f(x)$. To see this, consider $g(x)$ and $f(x)$ for arbitrary input x and let the traversal down the decision list terminate at a leaf on level l. This means $t_s = 0$ for all $s \in \{0, ..., l-2\}$. In particular, $|t_{l-1}| = 2^{n+1-l}$. Then, no matter the satisfiability of leaf edges level l and below, $\text{sgn}(\sum_{l=0}^{k-1} t_l + t_k^{(1)} + t_k^{(2)}) = \text{sgn}(t_l)$ since $\sum_{r=0}^{l} 2^r = 2^{r+1} - 1$. Therefore, $g(x)$ correctly reports the value of the first satisfied leaf edge in the decision list computing $f(x)$ and we can conclude that f is therefore a linear threshold function.

□

Problem 3.24 Consider □

Problem 3.25 Let f be computable by read-once decision tree T. Notice that T must be complete. For, if not, there exists an internal node v with fewer than two children. Suppose v lies at level l. For $l = k-1$, v lies at the deepest level and does not have two leaf children – a contradiction. For $l < k-1$, v has exactly one child which we denote w. But this means that any path reaching v must reach w, giving rise to an equivalent simplified tree T' with node v spliced out. Tree T', however, then has a leaf to node path via w of length less than k – a contradiction.

We show that for internal node v representing variable x_i at level $l \in \{0, ..., k-2\}$ we have $\mathbf{Inf}_i[f] = 2^{-l-1}$. To see this, first notice both the left and right subtrees of v contain equal amounts of -1 and 1 leaf nodes. We denote $X_v = \{x \in \{0,1\}^n | \text{The path of } x \text{ in T passes through } v\}$. Notice that $|X_v| = 2^{n-l}$. Now consider flipping coordinate i for an $x \in X_v$. Because T is read-once, the path x takes in the right subtree of v is independent of the path taken in the left subtree. Then, $\mathbb{P}_{x \in X_v}[f(x^{\oplus i}) = 1] = \mathbb{P}_{x \in X_v}[f(x^{\oplus i}) = -1] = \frac{1}{2}$. Therefore, $\mathbb{P}_{x \in X_v}[f(x^{\oplus i}) \neq f(x)] = \frac{1}{2}$. The influence of coordinate i is then $\frac{\frac{1}{2} \cdot 2^{n-l}}{2^n} = 2^{-l-1}$.

For internal node s representing coordinate j on level $l = k-1$ have influence $2^{n-l} = 2^{n-(k-1)}$. By a similar, argument, $2^{n-(k-1)}$ inputs x reach node s at the deepest level. Because each node at level k has a leaf of each -1 and 1, it follows that $\mathbb{P}_{x \in X_s}[f(x^{\oplus j}) \neq f(x)] = 1$. The influence of coordinate j is then $\frac{2^{n-k+1}}{2^n} = 2^{1-k}$.

The total influence is then

$$
\begin{aligned}
\mathbf{I}[f] &= 2^{-n} \left(2^{n-k+1} \cdot 2^{k-1} + \sum_{l=0}^{k-2} 2^{n-l-1} \cdot 2^l \right) && \text{(Completeness of T)} \\
&= 2^{-n} \left(2^n + (k-2) \cdot 2^{n-1} \right) \\
&= 1 + \frac{k-2}{2} = \frac{k}{2}
\end{aligned}
$$

This can also be seen via expected sensitivity. $\mathbf{I}[f] = \mathbb{E}_x[\mathbf{sens}(x)] = \frac{k}{2}$ since every input x traverses a path of length k in T, where the probability that any variable is pivotal is $\frac{1}{2}$ (due to the independence of subtrees and even leaf count). □

Problem 3.28 By definition, we have:

$$
\widehat{f^{\subseteq J}}(S) = \mathbb{E}_{x \sim \{-1,1\}^n}[f^{\subseteq J}(x) \cdot \chi_S(x)] = \mathbb{E}_{x \sim \{-1,1\}^n}[\mathbb{E}_{y \sim \{-1,1\}^{\overline{J}}}[f(x_J, y)] \cdot \chi_S(x)].
$$

Now, if $S \subseteq J$ then $\chi_S(x)$ depends only on x_J, so the above expression is equal to:

$$
\mathbb{E}_{x \sim \{-1,1\}^J}[\mathbb{E}_{y \sim \{-1,1\}^{\overline{J}}}[f(x_J, y)] \cdot \chi_S(x)] = \mathbb{E}_{z \in \{-1,1\}^n}[f(z) \cdot \chi_S(z)] = \widehat{f}(S).
$$

If instead $S \nsubseteq J$, then we can find some $i \in S \backslash J$. Then, we can rewrite the previous expression as:

$$
\mathbb{E}_{x_{\neq i} \in \{-1,1\}^{n-1}} \left[\mathbb{E}_{y \sim \{-1,1\}^{\overline{J}}} \left[\mathbb{E}_{x_i \sim \{-1,1\}}[f(x_J, y) \cdot \chi_S(x)] \right] \right]
$$

Since $i \in S$, we have $\chi_S(x) = -\chi_S(x^{\oplus i})$, and since $i \notin J$, we have $f(x_J, y) = f((x^{\oplus i})_J, y)$. Therefore, flipping x_i from 1 to -1 will negate the random variable in the innermost expectation, meaning the innermost expectation (over x_i) evaluates to 0, and hence $\widehat{f^{\subseteq J}}(S) = 0$ in this case. The desired Fourier expansion follows. □

Problem 3.30 Since the leaf b is a depth-k, the set of coordinates J encountered on the computation path to b satisfies $|J| \leq k$. Moreover, J has the property that $f(a, y) = b$ for any $a \in \{-1,1\}^J$, $y \in \{-1,1\}^{\overline{J}}$. It follows that $x_J = a \implies f^{\subseteq J}(x) = b$. Thus $f^{\subseteq J}(x) = \sum_{S \subseteq J} \widehat{f}(S) \chi_S(x) = b$. Let $x \in \{-1,1\}^n$ such that $x_J = a$. Then:

$$
\begin{aligned}
|b| &= \left| \sum_{S \subseteq J} \widehat{f}(S) \chi_S(x) \right| \\
&\leq \sum_{S \subseteq J} |\widehat{f}(S) \chi_S(x)| \\
&= \sum_{S \subseteq J} |\widehat{f}(S)|.
\end{aligned}
$$

Therefore, there exists some $S \subseteq J$ for which $|\widehat{f}(S)| \geq \frac{|b|}{2^{|J|}} \geq \frac{|b|}{2^k}$. Hence $\hat{\|} f \hat{\|}_\infty \geq \frac{|b|}{2^k}$, as desired. □

Problem 3.33 To learn f with 0 error, we must randomly sample $(x, f(x))$ for all x. We denote random variable X_i to be the number of draws required to obtain the ith unique pair, given that we have obtained $i-1$ unique pairs. Notice then that $X = \sum_{i=1}^{2^n} X_i$ is the total number of random draws needed to learn f with 0 error. In expectation

$$\mathbb{E}[X] = \mathbb{E}[\sum_{i=1}^{2^n} X_i] = \sum_{i=1}^{2^n} \mathbb{E}[X_i]$$

Notice that X_i is a geometric random variable with parameter $p = \frac{2^n-i+1}{2^n}$. Therefore, $\mathbb{E}[X_i] = \frac{2^n}{2^n-i+1}$ and

$$\begin{aligned} \mathbb{E}[X] &= 2^n \sum_{i=1}^{2^n} \frac{1}{2^n - i + 1} \\ &= 2^n \sum_{i=1}^{2^n} \frac{1}{i} \\ &\leq 2^n \log(2^n) \end{aligned}$$

Therefore, we require $\tilde{O}(2^n)$ draws in expectation to learn f with zero error. Then, by the Markov bound

$$\mathbb{P}[\sum_{i=1}^{2^n} X_i \geq 10n2^n] \leq \frac{n2^n}{10n2^n} = 0.1$$

So f can be learned with 0 error with 0.9 probability using $\tilde{O}(2^n)$ random samples. $\square$

Problem 3.40

(a) Our verification algorithm will work as follows: sample k (to be determined later) random examples $(x, f(x))$ and compute each $h(x)$. If the proportion of examples where $f(x) \neq h(x)$ is less than $\frac{3\varepsilon}{4}$ then output 'YES', and otherwise output 'NO'. Formally, for $1 \leq i \leq k$ let $X_i = 1$ if $f(x_i) \neq h(x_i)$ and 0 otherwise. Let $Y = \sum_{i=1}^{k} X_i$ and $\overline{X} := \mathbb{E}[Y] = \mathrm{dist}(f, h) \cdot k$. Observe that in order for our algorithm to mistakenly reject h for which $\mathrm{dist}(f, h) \leq \varepsilon/2$, or for it to mistakenly accept h for which $\mathrm{dist}(f, h) > \varepsilon$, we would need $|Y - \overline{X}| \geq \frac{\varepsilon k}{4}$. But by the Chernoff bound, we have:

$$\begin{aligned} \Pr\big[|Y - \overline{X}| \geq \varepsilon k/4\big] &\leq e^{-ck\varepsilon^2} \\ &= e^{-\log(1/\delta)} && (\text{set } k = \tfrac{\log(1/\delta)}{c\varepsilon^2}) \\ &= \delta. \end{aligned}$$

Hence, our algorithm works with the desired error upper bound. Now, since we request $k = \log(1/\delta) \cdot \mathrm{poly}(1/\varepsilon)$ examples, each of which taking $O(n+T)$ time (parsing x, computing h), in total our verification algorithm takes $\mathrm{poly}(n, T, 1/\varepsilon) \cdot \log(1/\delta)$ time, as desired.

(b) need to finish

$\square$

Problem 3.43

(a) Observe that

$$\mathbb{P}_x[\gamma \cdot x = \gamma' \cdot x] = \mathbb{P}_x[(\gamma \oplus \gamma') \cdot x = 0]$$

where we denote $\gamma'' \in \{0,1\}^n$. Since $\gamma \neq \gamma'$, there must exist some index $i \in [n]$ for which $\gamma''_i = 1$. Now consider all inputs $x \in \{0,1\}^{n-1}$ that map to indices $[n]\backslash\{i\}$. Then,

$$\gamma'' \cdot x = \left[\sum_{j \in [n]\backslash\{i\}} \gamma''_j x_j\right] + x_i$$

We establish a bijection between the set of x such that $\gamma'' \cdot x = 0$ and x such that $\gamma'' \cdot x = 1$. Let $x \in \{0,1\}^{n-1}$ correspond to the indices $[n]\backslash\{i\}$. Then, $c = \left[\sum_{j\in[n]\backslash\{i\}} \gamma''_j x_j\right] \in \{0,1\}$. Then, $\gamma'' \cdot x = c + x_i$ and the two settings of x_i yield different results to the dot product. We then have a 1-1 correspondence between the two sets and can therefore conclude that

$$\mathbb{P}_x[\gamma \cdot x = \gamma' \cdot x] = \frac{1}{2}$$

(b) We begin by observing that if $x^{(1)}, ..., x^{(m)}$ are the n basis vectors of $\widehat{\mathbb{F}_2^n}$ then

$$\begin{aligned} \forall i \in [n].\ \gamma \cdot x^{(i)} = \gamma' \cdot x^{(i)} &\iff \forall i \in [n].\ (\gamma \oplus \gamma') \cdot x^{(i)} = 0 \\ &\iff \gamma \oplus \gamma' \in \widehat{F_2^n}^{\perp} \iff \gamma \oplus \gamma' = 0 \iff \gamma = \gamma' \end{aligned}$$

So drawing n linearly independent vectors will guarantee the equivalence of γ and γ'. Given that we have drawn k linearly independent vectors, its span has size 2^k. Then, the probability of drawing a new linearly independent vector is $\frac{2^n-2^k}{2^n}$. Therefore, it takes $\frac{2^n}{2^n-2^k}$ random draws in expectation to find the $k+1$th new linearly independent vector. We denote random variable X_i to count the number of draws to find the ith linearly independent vector given $i-1$ have been found. By the Markov bound,

$$\begin{aligned} \mathbb{P}[X = \sum_{i=1}^{n} X_i \geq 20n] &\leq \frac{\mathbb{E}[X]}{20n} = \frac{1}{20n} \sum_{i=1}^{n} \frac{2^n}{2^n - 2^{i-1}} \\ &\leq \frac{2n}{20n} = \frac{1}{10} \end{aligned}$$

Therefore, for $C \geq 20$, $x^{(1)}, ..., x^{(m)}$ will contain a basis for $\widehat{\mathbb{F}_2^n}$ with at least 0.9 probability. Then $\forall i \in [n].\ \gamma \cdot x^{(i)} \iff \gamma = \gamma'$.

(c) Linear functions $f : \{0,1\}^n \to \{0,1\}$ are of the form $f(x) = a \cdot x$. Consider drawing $m = Cn$ random $(x, f(x))$ pairs for C computed in part (b). With high probability, a basis will exist

in this set. We construct matrix $M \in \{0,1\}^{n \times n}$ with row i equal to $x_b^{(i)}$, the ith basis vector, and vector v with $v_i = f(x_b^{(i)})$. We then solve the system

$$M\gamma = v$$

by computing $\gamma = vM^{-1}$. Notice that M has full rank since it comprises of a basis of $\widehat{\mathbb{F}_2^n}$ and is thus invertible. Matrix inversion and multiplication both take roughly $O(n^w)$ time.

$\square$

4 DNF Formulas and Small Depth Circuits

Problem 4.1 For a function $f : \{0,1\}^n \to \{0,1\}$, we construct the following DNF ϕ:

$$\phi = \bigvee_{(a_1,\ldots,a_n)\in f^{-1}(1)} \bigwedge_{i=1}^{n} \begin{cases} x_i & \text{if } a_i = 1 \\ \overline{x_i} & \text{if } a_i = 0 \end{cases}$$

By construction, we have $f(a_1, \ldots, a_n) = 1 \iff \phi(a_1, \ldots, a_n) = 1$, meaning $f = \phi$. Moreover, ϕ consists of at most 2^n terms, each of which has width n, as desired. $\square$

Problem 4.2 Consider CNF f given by

$$f = \bigwedge_{c \in C} \bigvee_{i \in c} x_i$$

where C is the set of clauses and each $c \in C$ is set of variable indices. Then notice that

$$\begin{aligned} f^{\dagger}(x) = \neg f(\neg x) &= \neg \bigwedge_{c\in C} \bigvee_{i \in c} \neg x_i \\ &= \bigvee_{c \in C} \bigwedge_{i \in c} x_i && (\text{DeMorgan's law}) \end{aligned}$$

which is exactly the DNF resulting from swapping ORs and ANDs. $\square$

Problem 4.3 If ϕ is a monotone DNF, we observe that flipping any coordinate of the input from False to True cannot make any term False, since the AND function is monotone. Therefore, it cannot turn the entire DNF formula from True to False, meaning ϕ computes a monotone function.

If $f : \{0,1\}^n \to \{0,1\}$ is a monotone function, we slightly adapt the construction from Problem 4.1:

$$\phi = \bigvee_{(a_1,\ldots,a_n)\in f^{-1}(1)} \bigwedge_{i=1}^{n} \begin{cases} x_i & \text{if } a_i = 1 \\ 1 & \text{if } a_i = 0 \end{cases}.$$

Here, the 1's in each term can be omitted, leaving a disjunction of conjunctions of unnegated variables and hence a monotone DNF ϕ. To show that ϕ computes f, first observe that $f(a) = 1$

Ryan O'Donnell Analysis of Boolean Functions Solutions

If you're looking for **solutions to exercises** from *Analysis of Boolean Functions* by Ryan O'Donnell, here's the honest situation:

📘 Official book (free PDF)

- You can access the full textbook here:
  👉 Read the book (PDF) ↗
- It's a graduate-level text covering Fourier analysis of Boolean functions, influences, noise stability, etc. arXiv

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❗ Are there official solutions?

- **No complete official solution manual is publicly released.**
- The book contains **hundreds of exercises**, but solutions are typically:
  - Used in **university courses**
  - Shared **partially** by instructors
  - Left as **research-style problems** for learning

✅ Available unofficial / partial solutions

There *are* some community or course-based resources:

1. Collaborative solution notes

- 👉 Solution Manual (student collaboration PDF) ↗
- Created by students (Spring 2023 reading group)
- Covers **selected exercises only**, not all
- Quality varies—use as guidance, not absolute authority

2. Course pages (with occasional solutions)

Some instructors post problem sets + solutions:

- Carnegie Mellon University course pages CMU School of C...
- Other universities (Chicago, Rutgers, etc.) sometimes release partial solutions or hints williamhoza.com

Thought for 35s ›

Let

$$f(x)=\sum_{S\subseteq[n]} a_S\,\chi_S(x)=\sum_{S\subseteq[n]} b_S\,\chi_S(x)$$

be two Fourier expansions of the same function $f:\{-1,1\}^n\to\mathbb{R}$, where $\chi_S(x)=\prod_{i\in S}x_i$. In O'Donnell's notation, this is the Fourier expansion on the Boolean cube, and Parseval's theorem says that for any $h:\{-1,1\}^n\to\mathbb{R}$,

$$\|h\|_2^2=\langle h,h\rangle=\sum_{S\subseteq[n]}\widehat{h}(S)^2.$$

Cambridge Assets +1

Now subtract the two expansions and define

$$h(x):=\sum_{S\subseteq[n]}(a_S-b_S)\chi_S(x).$$

Because the two sums both equal $f(x)$, we have $h(x)\equiv 0$. Hence

$$\|h\|_2^2=0.$$

But the Fourier coefficient of h on S is exactly a_S-b_S, so Parseval gives

$$0=\|h\|_2^2=\sum_{S\subseteq[n]}(a_S-b_S)^2.$$

CMU School of C... +1

A sum of squares can be 0 only if every term is 0. Therefore, for every $S\subseteq[n]$,

$$a_S-b_S=0,$$

so $a_S=b_S$ for all S. Thus the coefficients are uniquely determined, and the Fourier expansion is unique. Cambridge Assets

This is the clean Parseval proof: if two Fourier expansions existed, their difference would be the zero function, and Parseval forces all coefficient differences to vanish.

Sources

📚 Why solutions are scarce

This book is treated more like a **research training text**:

- Exercises are often **non-trivial proofs**
- Many problems build intuition for topics like:
  - Fourier expansion
  - Influence & noise sensitivity
  - Hypercontractivity
- Instructors intentionally avoid releasing full solutions to preserve learning value

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👍 What you can do instead

If you're stuck on specific problems, the best approach is:

- Work through **lecture notes** (many are online)
- Use **discussion forums / StackExchange**
- Ask for **step-by-step help** (I can do this)

👉 If you want

Tell me:

- Chapter + exercise number (e.g., "Chapter 2, Exercise 15")

I can walk you through a **complete solution or hint sequence**, tailored to your level.

Sources

Use Parseval's Theorem to prove uniqueness of the Fourier expansion

I'm checking a couple of reliable sources for the exact Fourier/Parseval setup Ryan O'Donnell uses, then I'll give a compact proof in that notation.

I found the exact ingredients: parity functions form an orthonormal basis, and Parseval/Plancherel gives equality of squared coefficients. I'm pulling the relevant pages so I can match the proof to the book's notation cleanly.

1.11 Suppose that $f:\{-1,1\}^n\to\{-1,1\}$ has $\deg(f)=k\ge 1$.
(*a*) Show that f's real multilinear representation over $\{0,1\}$ (see Exercise 1.9), call it $q(x)$, also has $\deg(q)=k$.
(*b*) Using Exercise 1.9(*c*),(*d*), deduce that f's Fourier spectrum is "2^{1-k}-granular", meaning each $\widehat{f}(S)$ is an integer multiple of 2^{1-k}.
(*c*) Show that $\sum_{S\subseteq[n]}|\widehat{f}(S)|\le 2^{k-1}$.

Thought for 2m 10s ›

Let

$$p(y)=f(y)=\sum_{S\subseteq[n]}\hat f(S)\chi_S(y),\qquad \chi_S(y)=\prod_{i\in S}y_i,$$

be the Fourier expansion of $f:\{-1,1\}^n\to\{-1,1\}$, and let $q(x)$ be the real multilinear representation of the same Boolean function over $\{0,1\}^n$. Exercise 1.9(d) gives the change of variables

$$q(x)=\frac12-\frac12\,p(1-2x_1,\dots,1-2x_n),$$

equivalently,

$$p(y)=1-2\,q\!\left(\frac{1-y_1}{2},\dots,\frac{1-y_n}{2}\right).$$

Exercise 1.9(c) says the coefficients of q are integers, and Exercise 1.11 is exactly the statement you screenshotted. Parseval says $\sum_S \hat f(S)^2=1$ for Boolean-valued f. CMU School of C... +1

For **(a)**, each multilinear monomial of degree d stays degree d under either substitution

$$y_i\mapsto 1-2x_i\quad\text{or}\quad x_i\mapsto\frac{1-y_i}{2},$$

because

$$\prod_{i\in T}(1-2x_i)\quad\text{and}\quad\prod_{i\in T}\frac{1-y_i}{2}$$

are both degree-$|T|$ polynomials. Hence from

$$q(x)=\frac12-\frac12\,p(1-2x)$$

we get $\deg(q)\le\deg(p)=k$, and from

$$p(y)=1-2\,q((1-y)/2)$$

we get $\deg(p) \le \deg(q)$. Therefore

$$\deg(q) = \deg(p) = k.$$

This proves part (a). CMU School of C...

For **(b)**, write

$$q(x) = \sum_{T \subseteq [n]} c_T \prod_{i \in T} x_i,$$

where by part (a) only terms with $|T| \le k$ appear, and by Exercise 1.9(c) each $c_T \in \mathbb{Z}$. Using

$$p(y) = 1 - 2\,q((1-y)/2),$$

we get

$$p(y) = 1 - 2\sum_T c_T \prod_{i \in T} \frac{1-y_i}{2} = 1 - \sum_T c_T\, 2^{1-|T|} \prod_{i \in T}(1-y_i).$$

Now expand

$$\prod_{i \in T}(1-y_i) = \sum_{S \subseteq T} (-1)^{|S|} \chi_S(y).$$

So every Fourier coefficient $\hat{f}(S)$ is an integer linear combination of numbers $2^{1-|T|}$ with $|T| \le k$. But each such number is an integer multiple of 2^{1-k}, since

$$2^{1-|T|} = 2^{1-k}\, 2^{k-|T|}.$$

Hence every $\hat{f}(S)$ is an integer multiple of 2^{1-k}. So the spectrum is 2^{1-k}-granular.
CMU School of C...

For **(c)**, let m be the number of nonzero Fourier coefficients. By part (b), each nonzero coefficient satisfies

$$|\hat{f}(S)| \ge 2^{1-k}.$$

Parseval gives

$$1 = \sum_S \hat{f}(S)^2 \ge m \cdot 2^{2-2k},$$

so

$$m \le 2^{2k-2}.$$

Now apply Cauchy–Schwarz:

$$\sum_S |\hat{f}(S)| \le \sqrt{m}\left(\sum_S \hat{f}(S)^2\right)^{1/2} \le \sqrt{2^{2k-2}} \cdot 1 = 2^{k-1}.$$

Thus

$$\sum_{S \subseteq [n]} |\hat{f}(S)| \le 2^{k-1}.$$

Done. CMU School of C... +1

If you want, I can also rewrite this as a polished homework-style proof with fewer intermediate steps.

Sources

1.19 Let $f : \{-1,1\}^n \to \{-1,1\}$.
(a) Suppose $\mathbf{W}^1[f] = 1$. Show that $f(x) = \pm\chi_S$ for some $|S| = 1$.
(b) Suppose $\mathbf{W}^{\le 1}[f] = 1$. Show that f depends on at most 1 input coordinate.
(c) Suppose $\mathbf{W}^{\le 2}[f] = 1$. Must f depend on at most 2 input coordinates? At most 3 input coordinates? What if we assume $\mathbf{W}^2[f] = 1$?

Solve part (c)

Thought for 6m 23s >

For part (c), the answer is **no in all three cases**.

In the book's notation, $W^k[f] = \sum_{|S|=k} \hat{f}(S)^2$, and for Boolean $f : \{-1,1\}^n \to \{-1,1\}$, Parseval says the total Fourier weight is 1. So $W^{\le 2}[f] = 1$ means exactly that all Fourier support is on levels $0, 1, 2$, i.e. f has Fourier degree at most 2. The exercise then asks whether that forces f to be a 2-junta or a 3-junta, and also asks about the stronger condition $W^2[f] = 1$. cs.cmu.edu +2

First, it does **not** force dependence on at most 2 coordinates. Consider

$$f_3(x_1,x_2,x_3) = \frac{x_1x_2 + x_1x_3 + x_2x_3 - 1}{2}.$$

Now

$$x_1x_2 + x_1x_3 + x_2x_3 = \begin{cases} 3, & \text{if } x_1 = x_2 = x_3, \\ -1, & \text{otherwise,} \end{cases}$$

so $f_3 \in \{-1,1\}$. Its polynomial has degree 2, hence $W^{\le 2}[f_3] = 1$. But it depends on all 3 variables; for example,

$$f_3(x_1,1,1) = x_1.$$

So the statement "must depend on at most 2 coordinates" is false.

It also does **not** force dependence on at most 3 coordinates. Consider

$$f_4(x_1,x_2,x_3,x_4) = \frac{-x_1x_2 - x_1x_3 - x_2x_4 + x_3x_4}{2}.$$

Rewrite it as

$$f_4(x) = \frac{x_3(x_4 - x_1) - x_2(x_1 + x_4)}{2}.$$

If $x_1 = x_4$, then $x_4 - x_1 = 0$, so

$$f_4(x) = -x_1x_2 \in \{-1,1\}.$$

If $x_1 = -x_4$, then $x_1 + x_4 = 0$, so

$$f_4(x) = -x_1x_3 \in \{-1,1\}.$$

Thus f_4 is Boolean-valued. It clearly has degree 2, and in fact it is homogeneous of degree 2.

Moreover, all four variables are relevant:

$$f_4(1,1,1,1) = -1, \qquad f_4(-1,1,1,1) = 1,$$

so x_1 matters;

$$f_4(1,1,1,1) = -1, \qquad f_4(1,-1,1,1) = 1,$$

so x_2 matters;

$$f_4(1,1,1,-1) = -1, \qquad f_4(1,1,-1,-1) = 1,$$

so x_3 matters;

$$f_4(-1,-1,1,-1) = -1, \qquad f_4(-1,-1,1,1) = 1,$$

so x_4 matters.

Since f_4 is Boolean-valued and has only degree-2 Fourier terms, all its Fourier weight lies on level 2; hence $W^2[f_4] = 1$. Therefore even the stronger assumption $W^2[f] = 1$ does **not** imply that f depends on at most 3 coordinates. cs.cmu.edu +1

So the conclusions are:

$$\boxed{\text{“At most 2 coordinates?” No.}}$$

$$\boxed{\text{“At most 3 coordinates?” No.}}$$

$$\boxed{\text{“If } W^2[f] = 1\text{?” Still no.}}$$

A clean final writeup would be: give f_3 as the counterexample to "at most 2", and f_4 as the counterexample to both "at most 3" and the stronger $W^2[f] = 1$ claim.

Sources

1.20 Let $f : \{-1,1\}^n \to \mathbb{R}$ satisfy $f = f^{=1}$. Show that $\mathbf{Var}[f^2] = 2\sum_{i \ne j} \hat{f}(i)^2 \hat{f}(j)^2$.

Thought for 1m 49s >

Here $f = f^{=1}$ means that f is equal to its level-1 Fourier part, so

$$f(x) = \sum_{i=1}^n \hat{f}(i)\, x_i.$$

This is exactly the notation O'Donnell uses for $f^{=k}$: $f^{=k} = \sum_{|S|=k} \hat{f}(S)\chi_S$, with $W^k[f] = \|f^{=k}\|_2^2$. cs.cmu.edu +1

Now square f:

$$f(x)^2 = \left(\sum_i \hat{f}(i)x_i\right)^2 = \sum_i \hat{f}(i)^2 x_i^2 + \sum_{i \ne j} \hat{f}(i)\hat{f}(j)x_ix_j.$$

Since $x_i^2 = 1$, this becomes

$$f(x)^2 = \sum_i \hat{f}(i)^2 + 2\sum_{i<j} \hat{f}(i)\hat{f}(j)x_ix_j.$$

Taking expectation, the mixed terms vanish because x_ix_j has mean 0 for $i \ne j$, so

$$\mathbb{E}[f^2] = \sum_i \hat{f}(i)^2.$$

Hence

$$f^2 - \mathbb{E}[f^2] = 2\sum_{i<j} \hat{f}(i)\hat{f}(j)x_ix_j.$$

The functions $x_i x_j$ are distinct Fourier characters, so by the Fourier covariance/Parseval identity, the variance is the sum of squares of these nonconstant coefficients.

cs.cmu.edu +1

Therefore,

$$\operatorname{Var}[f^2] = \left\|f^2 - \mathbb{E}[f^2]\right\|_2^2 = 4\sum_{i<j} \hat{f}(i)^2 \hat{f}(j)^2.$$

Finally,

$$4\sum_{i<j} \hat{f}(i)^2 \hat{f}(j)^2 = 2\sum_{i \neq j} \hat{f}(i)^2 \hat{f}(j)^2,$$

because the sum over ordered pairs $i \neq j$ counts each unordered pair twice. So

$$\boxed{\operatorname{Var}[f^2] = 2\sum_{i \neq j} \hat{f}(i)^2 \hat{f}(j)^2.}$$

cs.cmu.edu +1

Sources

1.28 (*a*) Generalize Exercise 1.5 as follows: Let $f : \mathbb{F}_2^n \to \{-1,1\}$ and suppose that $\operatorname{dist}(f, \chi_{S^*}) = \delta$. Show that $|\widehat{f}(S)| \le 2\delta$ for all $S \neq S^*$. (Hint: Use the union bound.)
(*b*) Deduce that the BLR Test rejects f with probability at least $3\delta - 10\delta^2 + 8\delta^3$.
(*c*) Show that this lower bound cannot be improved to $c\delta - O(\delta^2)$ for any $c > 3$.

Solve part (c)

Thought for 3m 22s >

Let $r(f)$ denote the BLR rejection probability. In the book's $\{\pm 1\}$-encoding, the BLR test accepts iff $f(x)f(y) = f(x+y)$; equivalently it rejects iff $f(x)f(y)f(x+y) = -1$. The book also remarks that for small ε, the BLR analysis can be improved to "more like $\varepsilon/3$-close to linear", and that this is sharp; Exercise 1.28(c) asks you to show exactly that sharpness. cs.cmu.edu +1

Here is a clean construction showing the coefficient 3 is best possible.

Fix a parity $\chi = \chi_{S^*}$. Let $N = 2^n$, choose an integer m, and pick a subset

$$A \subseteq \mathbb{F}_2^n$$

uniformly among all subsets of size m. Define

$$f(x) = \begin{cases} \chi(x), & x \notin A, \\ -\chi(x), & x \in A. \end{cases}$$

Then

$$\operatorname{dist}(f, \chi) = \frac{m}{N} =: \delta.$$

Now write $f = \chi g$, where

$$g(x) = \begin{cases} 1, & x \notin A, \\ -1, & x \in A. \end{cases}$$

Since $\chi(x)\chi(y) = \chi(x+y)$, the BLR predicate for f is the same as for g:

$$f(x)f(y) = f(x+y) \iff g(x)g(y) = g(x+y).$$

So we only need to analyze g.

For a test triple $(x, y, x+y)$, the test rejects exactly when an odd number of the three points lie in A. Indeed,

$$g(x)g(y)g(x+y) = -1$$

iff among $x, y, x+y$ there are 1 or 3 negative signs.

Assume first that $x, y, x+y$ are distinct. Then, because A is a uniform m-subset of an N-point set,

$$\Pr[\text{reject} \mid x, y, x+y \text{ distinct}] = \frac{\binom{m}{1}\binom{N-m}{2} + \binom{m}{3}}{\binom{N}{3}}.$$

With $\delta = m/N$, this is

$$3\delta(1-\delta)^2 + \delta^3 + O\left(\frac{1}{N}\right) = 3\delta - 6\delta^2 + 4\delta^3 + O\left(\frac{1}{N}\right).$$

The triple fails to be distinct only if $x = 0$, or $y = 0$, or $x = y$, which has probability at most $3/N$. Hence

$$\mathbb{E}_A[r(f)] \le 3\delta - 6\delta^2 + 4\delta^3 + O\left(\frac{1}{N}\right).$$

Therefore for some choice of A,

$$r(f) \le 3\delta - 6\delta^2 + 4\delta^3 + O\left(\frac{1}{N}\right).$$

Now choose n large enough that $1/N = O(\delta^2)$ and, say, $\delta = m/N$ is dyadic. Then we get an explicit Boolean function f with

$$\operatorname{dist}(f, \chi_{S^*}) = \delta \quad \text{and} \quad r(f) \le 3\delta - O(\delta^2).$$

This disproves any universal lower bound of the form

$$r(f) \ge c\delta - O(\delta^2)$$

with $c > 3$: for sufficiently small δ, the right-hand side would be strictly larger than $3\delta - O(\delta^2)$, while the construction above has rejection probability at most $3\delta - O(\delta^2)$.

So the answer to part (c) is:

$$\boxed{\text{The coefficient 3 is best possible; it cannot be replaced by any } c > 3.}$$

If you want, I can also turn this into a shorter "homework submission" version with the algebra compressed.

Sources

2.5 Let $f : \{-1,1\}^n \to \{-1,1\}$ and $i \in [n]$. We say that f is *unate in the ith direction* if either $f(x^{(i\mapsto -1)}) \le f(x^{(i\mapsto 1)})$ for all x (*monotone in the ith direction*) or $f(x^{(i\mapsto -1)}) \ge f(x^{(i\mapsto 1)})$ for all x (*antimonotone in the ith direction*). We say that f is *unate* if it is unate in all n directions.
(*a*) Show that $|\widehat{f}(i)| \le \mathbf{Inf}_i[f]$ with equality if and only if f is unate in the ith direction.
(*b*) Show that the second statement of Theorem 2.33 holds even for all unate f.

2.6 Show that linear threshold functions are unate.

Thought for 2m 15s >

Here are concise solutions to **2.5** and **2.6**.

For context, the book defines $D_i f$ so that for Boolean-valued f, $D_i f(x) \in \{-1, 0, 1\}$, and it gives

$$\backslash\mathrm{Inf}_i[f] = \mathbb{E}[(D_i f)^2], \qquad D_i f(x) = \sum_{S \ni i} \hat{f}(S) x_{S \setminus \{i\}}.$$

Hence $\mathbb{E}[D_i f] = \hat{f}(i)$. Theorem 2.33's second statement is that for monotone Boolean f,

$$I[f] \le I[\mathrm{Maj}_n] = \sqrt{2/\pi}\,\sqrt{n} + O(n^{-1/2}).$$

CMU School of C... +2

2.5(a)

Let

$$X := D_i f(x).$$

Since f is Boolean-valued, $X \in \{-1, 0, 1\}$. Therefore

$$|X| = X^2.$$

Using the identities above,

$$|\hat{f}(i)| = |\mathbb{E}[X]| \le \mathbb{E}[|X|] = \mathbb{E}[X^2] = \backslash\mathrm{Inf}_i[f].$$

So

$$|\hat{f}(i)| \le \backslash\mathrm{Inf}_i[f].$$

CMU School of C... +1

Equality holds iff

$$|\mathbb{E}[X]| = \mathbb{E}[|X|],$$

which for a real random variable means that X has a fixed sign almost surely. Since $X \in \{-1, 0, 1\}$, this is equivalent to either $D_i f(x) \ge 0$ for all x, or $D_i f(x) \le 0$ for all x. But $D_i f(x) \ge 0$ is exactly

$$f(x^{(i\to -1)}) \le f(x^{(i\to 1)}),$$

and $D_i f(x) \le 0$ is exactly the reverse inequality. Thus equality holds iff f is unate in the ith direction. CMU School of C...

2.5(b)

Assume f is unate. For each coordinate i, choose $\sigma_i \in \{\pm 1\}$ so that after flipping x_i when necessary, the function becomes monotone in that coordinate. Define

$$g(x_1, \dots, x_n) = f(\sigma_1 x_1, \dots, \sigma_n x_n).$$

Then g is monotone.

Flipping input signs is just a relabeling of the cube, so it preserves each coordinate influence; hence

$$\Inf_i[g] = \Inf_i[f] \quad \text{for all } i, \qquad \text{and therefore} \qquad I[g] = I[f].$$

Now apply the second statement of Theorem 2.33 to the monotone function g:

$$I[f] = I[g] \le I[\mathrm{Maj}_n].$$

So the same bound holds for every unate f:

$$I[f] \le I[\mathrm{Maj}_n] = \sqrt{2/\pi}\,\sqrt{n} + O(n^{-1/2}).$$

2.6

Let

$$f(x) = \mathrm{sgn}(a_0 + a_1 x_1 + \cdots + a_n x_n)$$

be a linear threshold function. Fix a coordinate i, and fix all other variables. Then f becomes a one-variable function

$$t \mapsto \mathrm{sgn}(b + a_i t), \qquad t \in \{-1, 1\},$$

where

$$b = a_0 + \sum_{j \ne i} a_j x_j.$$

So

$$f(x^{(i \to -1)}) = \mathrm{sgn}(b - a_i), \qquad f(x^{(i \to 1)}) = \mathrm{sgn}(b + a_i).$$

If $a_i \ge 0$, then $b - a_i \le b + a_i$, hence

$$\mathrm{sgn}(b - a_i) \le \mathrm{sgn}(b + a_i),$$

so f is monotone in direction i.

If $a_i \le 0$, then $b - a_i \ge b + a_i$, hence

$$\mathrm{sgn}(b - a_i) \ge \mathrm{sgn}(b + a_i),$$

so f is antimonotone in direction i.

Thus every coordinate is either monotone or antimonotone, meaning every linear threshold function is unate.

If you want, I can also turn this into a tighter "submitted-solutions" version with just the core lines of proof.

2.8 Let $f : \{-1,1\}^n \to \{-1,1\}$. Without using Fourier formulas, show that $\mathbf{Inf}_i[f] \le \mathbf{Var}[f]$ for each $i \in [n]$. (Hint: Show $\mathbf{Inf}_i[f] \le 2\min\{\mathbf{Pr}[f = -1], \mathbf{Pr}[f = 1]\}$.)

2.9 Let $f : \{0,1\}^6 \to \{-1,1\}$ be given by the weighted majority $f(x) = \mathrm{sgn}(-58 + 31x_1 + 31x_2 + 28x_3 + 21x_4 + 2x_5 + 2x_6)$. Compute $\mathbf{Inf}_i[f]$ for all $i \in [6]$.

2.10 Given $b \in \{-1,1\}$, say that coordinate i is *b-pivotal* for $f : \{-1,1\}^n \to \{-1,1\}$ on input x if $f(x) = b$ and $f(x^{\oplus i}) \ne b$. Show that $\mathbf{Pr}_{\boldsymbol{x}}[i \text{ is } b\text{-pivotal on } \boldsymbol{x}] = \frac{1}{2}\mathbf{Inf}_i[f]$. Deduce that $\mathbf{I}[f] = 2\mathbf{E}_{\boldsymbol{x}}[\#\ b\text{-pivotal coordinates on } \boldsymbol{x}]$.

Thought for 4m 1s

Here are clean solutions to **2.8, 2.9, and 2.10**. I'll use the standard combinatorial meaning of influence for Boolean-valued functions: $\Inf_i[f]$ is the probability that flipping bit i changes the value of f, and $I[f] = \sum_i \Inf_i[f]$. For Boolean f, the book also gives $\Var[f] = 4\Pr[f=1]\Pr[f=-1]$.

2.8

Let

$$A = \{x : f(x) = 1\}, \qquad B = \{x : f(x) = -1\},$$

and write

$$p = \Pr[f=1] = \frac{|A|}{2^n}, \qquad q = \Pr[f=-1] = \frac{|B|}{2^n}.$$

Fix a coordinate i. Consider the 2^{n-1} undirected i-edges of the cube, i.e. pairs $\{x, x^{\oplus i}\}$. An i-edge contributes to $\Inf_i[f]$ exactly when one endpoint lies in A and the other in B. Hence

$$\Inf_i[f] = \frac{\#\{\text{bichromatic } i\text{-edges}\}}{2^{n-1}}.$$

Now each vertex lies on exactly one i-edge, so the number of bichromatic i-edges is at most the size of the smaller side:

$$\#\{\text{bichromatic } i\text{-edges}\} \le \min(|A|, |B|).$$

Therefore

$$\Inf_i[f] \le \frac{\min(|A|,|B|)}{2^{n-1}} = 2\min(p, q).$$

Finally,

$$\Var[f] = 4pq = 2\min(p,q) \cdot 2\max(p,q) \ge 2\min(p,q),$$

since $\max(p,q) \ge \frac{1}{2}$. Thus

$$\boxed{\Inf_i[f] \le \Var[f].}$$

This proves Exercise 2.8.

2.9

Let

$$f(x) = \mathrm{sgn}(-58 + 31x_1 + 31x_2 + 28x_3 + 21x_4 + 2x_5 + 2x_6), \qquad x \in \{0,1\}^6.$$

No subset of the weights sums to 58, so there are no tie cases. For a fixed coordinate i, flipping x_i changes the output iff the sum of the *other* weights lies strictly between $58 - w_i$ and 58.

For $i = 1$ (and symmetrically $i = 2$), the other-weight sum is

$$s = 31x_2 + 28x_3 + 21x_4 + 2x_5 + 2x_6.$$

We need $27 < s < 58$. This happens exactly in the two cases

$$(x_2, x_3) = (1, 0) \quad \text{or} \quad (0, 1),$$

and then x_4, x_5, x_6 are arbitrary. So there are

$$2 \cdot 2^3 = 16$$

good assignments out of $2^5 = 32$, giving

$$\Inf_1[f] = \Inf_2[f] = \frac{16}{32} = \frac{1}{2}.$$

For $i = 3$, the other-weight sum is

$$s = 31x_1 + 31x_2 + 21x_4 + 2x_5 + 2x_6,$$

and we need $30 < s < 58$. This happens exactly when **exactly one** of x_1, x_2 equals 1; then x_4, x_5, x_6 are arbitrary. Again this gives 16 good assignments out of 32, so

$$\Inf_3[f] = \frac{1}{2}.$$

For $i = 4$, the other-weight sum is

$$s = 31x_1 + 31x_2 + 28x_3 + 2x_5 + 2x_6,$$

and we would need $37 < s < 58$. But if at most one of x_1, x_2, x_3 is 1, then $s \le 35$; if $x_3 = 1$ and one of x_1, x_2 is 1, then $s \ge 59$. So this is impossible, and

$$\Inf_4[f] = 0.$$

For $i = 5$ (and symmetrically $i = 6$), the other-weight sum is

$$s = 31x_1 + 31x_2 + 28x_3 + 21x_4 + 2x_6,$$

and we would need $56 < s < 58$, i.e. $s = 57$. But 57 is not attainable from these weights, so

$$\Inf_5[f] = \Inf_6[f] = 0.$$

Hence

$$\boxed{\Inf_1[f] = \Inf_2[f] = \Inf_3[f] = \frac{1}{2}, \qquad \Inf_4[f] = \Inf_5[f] = \Inf_6[f] = 0.}$$

The exercise statement is exactly this weighted-majority example.

2.10

Fix $b \in \{-1, 1\}$. Define

$$B_i^b(x) = \mathbf{1}[f(x) = b \text{ and } f(x^{\oplus i}) \ne b].$$

Then $B_i^b(x) = 1$ exactly when i is b-pivotal on x.

Now on any input x,

$$\mathbf{1}[f(x) \ne f(x^{\oplus i})] = B_i^b(x) + B_i^b(x^{\oplus i}),$$

because if the i-edge is bichromatic, exactly one endpoint has value b. Taking expectations over uniform x,

$$\Inf_i[f] = \Pr[f(x) \ne f(x^{\oplus i})] = \mathbb{E}[B_i^b(x)] + \mathbb{E}[B_i^b(x^{\oplus i})].$$

Since $x \mapsto x^{\oplus i}$ is a bijection of the cube, the two expectations are equal. Therefore

$$\Inf_i[f] = 2\,\mathbb{E}[B_i^b(x)] = 2\,\Pr[i \text{ is } b\text{-pivotal on } x].$$

So

$$\boxed{\Pr_x[i \text{ is } b\text{-pivotal on } x] = \frac{1}{2}\Inf_i[f].}$$

Summing over i and using linearity of expectation,

$$I[f] = \sum_{i=1}^{n} \textcolor{red}{\backslash\text{Inf}}_i[f] = 2\sum_{i=1}^{n} \Pr[i \text{ is } b\text{-pivotal on } x] = 2\,\mathbb{E}_x\left[\sum_{i=1}^{n} \mathbf{1}(i \text{ is } b\text{-pivotal on } x)\right]$$

Thus

$$\boxed{I[f] = 2\,\mathbb{E}_x[\#\ b\text{-pivotal coordinates on } x].}$$

This matches the book's definition of total influence and the exercise statement.

CMU School of C... +1

If you want, I can turn these into a tighter "submission-style" version with just the final polished proofs.

Sources

2.14 Let $f : \{-1,1\}^n \to \mathbb{R}$, and write $g = |f|$. Show that $|\mathrm{D}_i g| \le |\mathrm{D}_i f|$ pointwise. Deduce that $\mathbf{Inf}_i[g] \le \mathbf{Inf}_i[f]$ and $\mathbf{I}[g] \le \mathbf{I}[f]$.

Thought for 35s >

Let

$$a = f(x^{(i\to 1)}), \qquad b = f(x^{(i\to -1)}).$$

By the book's definition,

$$D_i f(x) = \frac{a-b}{2}, \qquad D_i g(x) = \frac{|a|-|b|}{2},$$

for $g = |f|$, and $\textcolor{red}{\backslash\text{Inf}}_i[h] = \mathbb{E}[(D_i h)^2]$ for any $h : \{-1,1\}^n \to \mathbb{R}$. CMU School of C... +1

Now use the elementary real inequality

$$\big||a|-|b|\big| \le |a-b|.$$

Therefore,

$$|D_i g(x)| = \frac{\big||a|-|b|\big|}{2} \le \frac{|a-b|}{2} = |D_i f(x)|.$$

So we have the desired pointwise bound

$$\boxed{|D_i g(x)| \le |D_i f(x)| \quad \text{for all } x.}$$

Squaring and taking expectation gives

$$\textcolor{red}{\backslash\text{Inf}}_i[g] = \mathbb{E}\big[(D_i g)^2\big] \le \mathbb{E}\big[(D_i f)^2\big] = \textcolor{red}{\backslash\text{Inf}}_i[f].$$

Hence

$$\boxed{\textcolor{red}{\backslash\text{Inf}}_i[g] \le \textcolor{red}{\backslash\text{Inf}}_i[f] \quad \text{for every } i.}$$

Using $I[h] = \sum_{i=1}^n \textcolor{red}{\backslash\text{Inf}}_i[h]$, we conclude

$$I[g] = \sum_{i=1}^{n} \textcolor{red}{\backslash\text{Inf}}_i[g] \le \sum_{i=1}^{n} \textcolor{red}{\backslash\text{Inf}}_i[f] = I[f].$$

So

$$\boxed{I[g] \le I[f].}$$

The identity $I[h] = \sum_i \textcolor{red}{\backslash\text{Inf}}_i[h]$ is the standard definition of total influence in this section. CMU School of C...

A compact submission version is:

$$D_i|f|(x) = \frac{|f(x^{(i\to 1)})| - |f(x^{(i\to -1)})|}{2},$$

so by $\big||u|-|v|\big| \le |u-v|$,

$$|D_i|f|(x)| \le \frac{|f(x^{(i\to 1)}) - f(x^{(i\to -1)})|}{2} = |D_i f(x)|.$$

Thus $\textcolor{red}{\backslash\text{Inf}}_i[|f|] = \mathbb{E}[(D_i|f|)^2] \le \mathbb{E}[(D_i f)^2] = \textcolor{red}{\backslash\text{Inf}}_i[f]$, and summing over i yields $I[|f|] \le I[f]$.

Sources

2.18 Let $f : \{-1,1\}^n \to \mathbb{R}$. Show that

$$\mathrm{L}f(x) = \frac{d}{d\rho}\mathrm{T}_\rho f(x)\Big|_{\rho=1} = -\frac{d}{dt}\mathrm{T}_{e^{-t}} f(x)\Big|_{t=0}.$$

2.19 Suppose $f, g : \{-1,1\}^n \to \mathbb{R}$ have the property that f does not depend on the ith coordinate and g does not depend on the jth coordinate ($i \ne j$). Show that $\mathbf{E}[\boldsymbol{x}_i \boldsymbol{x}_j f(\boldsymbol{x}) g(\boldsymbol{x})] = \mathbf{E}[\mathrm{D}_j f(\boldsymbol{x}) \mathrm{D}_i g(\boldsymbol{x})]$.

Thought for 1m 41s >

Here are clean proofs of both exercises.

For **2.18**, use the Fourier formulas

$$T_\rho f = \sum_{S\subseteq[n]} \rho^{|S|} \hat{f}(S)\chi_S \quad \text{and} \quad Lf = \sum_{S\subseteq[n]} |S|\,\hat{f}(S)\chi_S.$$

These are exactly Proposition 2.47 and Proposition 2.37 in O'Donnell's notation.

CMU School of C... +1

Because the sum is finite, we can differentiate term-by-term:

$$\frac{d}{d\rho} T_\rho f(x) = \sum_{S\subseteq[n]} |S|\,\rho^{|S|-1} \hat{f}(S)\chi_S(x).$$

Evaluating at $\rho = 1$ gives

$$\frac{d}{d\rho} T_\rho f(x)\Big|_{\rho=1} = \sum_{S\subseteq[n]} |S|\,\hat{f}(S)\chi_S(x) = Lf(x).$$

So

$$Lf(x) = \frac{d}{d\rho} T_\rho f(x)\Big|_{\rho=1}.$$

This proves the first identity. CMU School of C... +1

Now write $\rho = e^{-t}$. Then

$$T_{e^{-t}} f = \sum_{S\subseteq[n]} e^{-t|S|} \hat{f}(S)\chi_S.$$

Differentiate with respect to t:

$$\frac{d}{dt} T_{e^{-t}} f(x) = -\sum_{S\subseteq[n]} |S|\,e^{-t|S|} \hat{f}(S)\chi_S(x).$$

Evaluating at $t = 0$,

$$\frac{d}{dt} T_{e^{-t}} f(x)\Big|_{t=0} = -\sum_{S\subseteq[n]} |S|\,\hat{f}(S)\chi_S(x) = -Lf(x).$$

Hence

$$Lf(x) = -\frac{d}{dt} T_{e^{-t}} f(x)\Big|_{t=0}.$$

So Exercise 2.18 is proved. CMU School of C... +1

For **2.19**, use the decomposition

$$h = x_k D_k h + E_k h$$

from Proposition 2.24, where neither $D_k h$ nor $E_k h$ depends on x_k. Also, if a function does not depend on the kth coordinate, then it equals its E_k-part. CMU School of C...

Since f does not depend on x_i, we have $f = E_i f$. Applying the j-decomposition to f,

$$f = x_j D_j f + E_j f.$$

Similarly, since g does not depend on x_j, we have $g = E_j g$, and applying the i-decomposition to g,

$$g = x_i D_i g + E_i g.$$

Therefore

$$x_i x_j f g = x_i x_j (x_j D_j f + E_j f)(x_i D_i g + E_i g).$$

Expanding,

$$x_i x_j f g = D_j f\, D_i g + x_i D_j f\, E_i g + x_j E_j f\, D_i g + x_i x_j E_j f\, E_i g.$$

This uses only Proposition 2.24. CMU School of C...

Now take expectation. The first term stays as is. The other three terms have expectation 0: in

$$\mathbb{E}[x_i D_j f\, E_i g],$$

the factor $D_j f\, E_i g$ does not depend on x_i, so averaging over x_i kills the term. Likewise,

$$\mathbb{E}[x_j E_j f\, D_i g] = 0$$

because $E_j f\, D_i g$ does not depend on x_j. And

$$\mathbb{E}[x_i x_j E_j f\, E_i g] = 0$$

because $E_i g$ is independent of x_i, so averaging over x_i already makes it vanish. Hence

$$\mathbb{E}[x_i x_j f(x) g(x)] = \mathbb{E}[D_j f(x) D_i g(x)].$$

This proves Exercise 2.19. CMU School of C...

Final answers:

$$\boxed{Lf(x) = \frac{d}{d\rho} T_\rho f(x)\Big|_{\rho=1} = -\frac{d}{dt} T_{e^{-t}} f(x)\Big|_{t=0}}$$

and

$$\boxed{\mathbb{E}[x_i x_j f(x) g(x)] = \mathbb{E}[D_j f(x)\, D_i g(x)]}.$$

Sources

2.21 Let $f : \{-1,1\}^n \to \mathbb{R}$ and $i \in [n]$.
(a) Define $\mathrm{Var}_i f : \{-1,1\}^n \to \mathbb{R}$ by

$$\mathrm{Var}_i f(x) = \underset{\boldsymbol{x}_i}{\mathbf{Var}}[f(x_1, \ldots, x_{i-1}, \boldsymbol{x}_i, x_{i+1}, \ldots, x_n)].$$

Show that $\mathbf{Inf}_i[f] = \mathbf{E}_{\boldsymbol{x}}[\mathrm{Var}_i f(\boldsymbol{x})]$.
(b) Show that

$$\mathbf{Inf}_i[f] = \tfrac{1}{2} \underset{\substack{\boldsymbol{x}_i, \boldsymbol{x}'_i \sim \{-1,1\} \\ \text{independent}}}{\mathbf{E}} \left[\left\| f_{|\boldsymbol{x}_i} - f_{|\boldsymbol{x}'_i} \right\|_2^2 \right],$$

where $f_{|b}$ denotes the function of $n-1$ variables gotten by fixing the ith input of f to bit b.

Thought for 49s ›

Let

$$D_i f(x) = \frac{f(x^{(i\to 1)}) - f(x^{(i\to -1)})}{2}, \qquad \backslash\mathrm{Inf}_i[f] = \mathbb{E}[(D_i f(x))^2].$$

These are the definitions used in §2.2 of O'Donnell. CMU School of C... +1

For **(a)**, fix all coordinates except x_i, and write

$$A = f(x^{(i\to 1)}), \qquad B = f(x^{(i\to -1)}).$$

Then, by definition,

$$\backslash\mathrm{Var}_i f(x) = \backslash\mathrm{Var}_{x_i}[f(x)] = \frac{A^2 + B^2}{2} - \left(\frac{A+B}{2}\right)^2 = \frac{(A-B)^2}{4}.$$

But

$$D_i f(x) = \frac{A-B}{2},$$

so

$$\backslash\mathrm{Var}_i f(x) = (D_i f(x))^2.$$

Now average over x:

$$\mathbb{E}_x[\backslash\mathrm{Var}_i f(x)] = \mathbb{E}_x[(D_i f(x))^2] = \backslash\mathrm{Inf}_i[f].$$

Hence

$$\boxed{\backslash\mathrm{Inf}_i[f] = \mathbb{E}_x[\backslash\mathrm{Var}_i f(x)].}$$

For **(b)**, let $h_+ = f|_{x_i=1}$ and $h_- = f|_{x_i=-1}$, viewed as functions on $\{-1,1\}^{n-1}$. For any $y \in \{-1,1\}^{n-1}$,

$$D_i f(y) = \frac{h_+(y) - h_-(y)}{2},$$

so from the definition of influence,

$$\backslash\mathrm{Inf}_i[f] = \mathbb{E}_y\left[\left(\frac{h_+(y) - h_-(y)}{2}\right)^2\right] = \frac{1}{4}\|h_+ - h_-\|_2^2.$$

On the other hand, if x_i, x'_i are independent uniform bits in $\{-1,1\}$, then

$$\|f|_{x_i} - f|_{x'_i}\|_2^2$$

equals 0 when $x_i = x'_i$, and equals $\|h_+ - h_-\|_2^2$ when $x_i \neq x'_i$. Since these each happen with probability $1/2$,

$$\mathbb{E}_{x_i, x'_i}\left[\|f|_{x_i} - f|_{x'_i}\|_2^2\right] = \frac{1}{2}\|h_+ - h_-\|_2^2.$$

Therefore,

$$\frac{1}{2}\mathbb{E}_{x_i, x'_i}\left[\|f|_{x_i} - f|_{x'_i}\|_2^2\right] = \frac{1}{4}\|h_+ - h_-\|_2^2 = \backslash\mathrm{Inf}_i[f].$$

So

$$\boxed{\backslash\mathrm{Inf}_i[f] = \frac{1}{2}\mathbb{E}_{x_i, x'_i \sim \{-1,1\}\ \text{indep.}}\left[\|f|_{x_i} - f|_{x'_i}\|_2^2\right].}$$

A compact submission version is:

$$\backslash\mathrm{Var}_i f(x) = \frac{f(x^{(i\to 1)})^2 + f(x^{(i\to -1)})^2}{2} - \left(\frac{f(x^{(i\to 1)}) + f(x^{(i\to -1)})}{2}\right)^2 = (D_i f(x))^2,$$

hence $\backslash\mathrm{Inf}_i[f] = \mathbb{E}_x[\backslash\mathrm{Var}_i f(x)]$. Also,

$$\backslash\mathrm{Inf}_i[f] = \frac{1}{4}\|f|_1 - f|_{-1}\|_2^2,$$

while for independent x_i, x'_i, the random quantity $\|f|_{x_i} - f|_{x'_i}\|_2^2$ is 0 with probability $1/2$ and $\|f|_1 - f|_{-1}\|_2^2$ with probability $1/2$, giving the claimed formula.

CMU School of C... +1

Sources

2.24 Prove Proposition 2.58 with $O(n^{-1})$ in place of $o_n(1)$. (Hint: Show $\widehat{f}(i) \le \frac{\sqrt{2/\pi}}{\sqrt{n}} + O(n^{-3/2})$ using Theorem 2.33.)
2.25 Deduce $\mathrm{T}_\rho f(x) = \sum_S \rho^{|S|} \widehat{f}(S) x^S$ using Exercise 1.4.

Thought for 2m 32s ›

Here are short proofs of both.

2.24

Let

$$\hat{f}(1) = \hat{f}(2) = \cdots = \hat{f}(n) = a.$$

Proposition 2.58 says that under this hypothesis one has $W_1[f] \le 2/\pi + o_n(1)$, and Exercise 2.24 asks for the sharper $O(n^{-1})$ error. The hint points to Theorem 2.33, which states that among all Boolean f, the quantity $\sum_{i=1}^n \hat{f}(i)$ is maximized by majority, and that

$$I[\mathrm{Maj}_n] = \sqrt{\frac{2}{\pi}}\sqrt{n} + O(n^{-1/2}).$$

CMU School of C... +1

Apply Theorem 2.33 to f. Since $\sum_i \hat{f}(i) = na$,

$$na = \sum_{i=1}^n \hat{f}(i) \le I[\mathrm{Maj}_n] = \sqrt{\frac{2}{\pi}}\sqrt{n} + O(n^{-1/2}).$$

Applying the same theorem to $-f$ gives

$$-na = \sum_{i=1}^n \widehat{(-f)}(i) \le I[\mathrm{Maj}_n].$$

Hence

$$|a| \le \frac{I[\mathrm{Maj}_n]}{n} = \sqrt{\frac{2}{\pi}}\frac{1}{\sqrt{n}} + O(n^{-3/2}).$$

This is exactly the estimate suggested in the hint. CMU School of C... +1

Now

$$W_1[f] = \sum_{i=1}^n \hat{f}(i)^2 = na^2,$$

so

$$W_1[f] \le n\left(\sqrt{\frac{2}{\pi}}\frac{1}{\sqrt{n}} + O(n^{-3/2})\right)^2 = \frac{2}{\pi} + O(n^{-1}).$$

Thus Proposition 2.58 can be sharpened to

$$\boxed{W_1[f] \le \frac{2}{\pi} + O(n^{-1}).}$$

CMU School of C... +1

2.25

Write

$$f(z) = \sum_{S \subseteq [n]} \hat{f}(S)\, z^S, \qquad z^S := \prod_{i \in S} z_i.$$

By definition,

$$T_\rho f(x) = \mathbb{E}_{y \sim N_\rho(x)}[f(y)],$$

where the coordinates y_i are independent and satisfy $\mathbb{E}[y_i] = \rho x_i$. Exercise 1.4 says that if $\mu \in [-1,1]^n$, then evaluating the multilinear polynomial f at μ equals the expectation of $f(y)$ when the bits y_i are independent with means μ_i. Therefore, with $\mu_i = \rho x_i$,

$$T_\rho f(x) = f(\rho x_1, \ldots, \rho x_n).$$

CMU School of C... +1

Substitute the multilinear expansion:

$$T_\rho f(x) = f(\rho x_1, \ldots, \rho x_n) = \sum_{S \subseteq [n]} \hat{f}(S) \prod_{i \in S}(\rho x_i) = \sum_{S \subseteq [n]} \rho^{|S|} \hat{f}(S) x^S.$$

So the desired formula is

$$\boxed{T_\rho f(x) = \sum_{S \subseteq [n]} \rho^{|S|} \hat{f}(S) x^S.}$$

CMU School of C... +1

This is exactly Proposition 2.47, but Exercise 2.25 derives it directly from Exercise 1.4.

CMU School of C...

Sources

2.50 Compute the precise probability of a Condorcet winner (under impartial culture) in a 3-candidate, 3-voter election using $f = \mathrm{Maj}_3$.

2.51 (a) Arrow's Theorem for 3 candidates is slightly more general than what we stated: it allows for three *different* unanimous functions $f, g, h : \{-1,1\}^n \to \{-1,1\}$ to be used in the three pairwise elections. But show that if using f, g, h always gives rise to a Condorcet winner then $f = g = h$. (Hint: First show $g(x) = -f(-x)$ for all x by using the fact that x, $y = -x$, and $z = (f(x), \dots, f(x))$ is always a valid possibility for the votes.)

(b) Extend Arrow's Theorem to the case of Condorcet elections with more than 3 candidates.

2.50 Compute the precise probability of a Condorcet winner (under impartial culture) in a 3-candidate, 3-voter election using $f = \mathrm{Maj}_3$.

2.51 (a) Arrow's Theorem for 3 candidates is slightly more general than what we stated: it allows for three *different* unanimous functions $f, g, h : \{-1,1\}^n \to \{-1,1\}$ to be used in the three pairwise elections. But show that if using f, g, h always gives rise to a Condorcet winner then $f = g = h$. (Hint: First show $g(x) = -f(-x)$ for all x by using the fact that x, $y = -x$, and $z = (f(x), \dots, f(x))$ is always a valid possibility for the votes.)

(b) Extend Arrow's Theorem to the case of Condorcet elections with more than 3 candidates.

Thought for 2m 29s >

Here are clean solutions to **2.50** and **2.51**.

2.50

For a 3-candidate election using a single Boolean rule f, Theorem 2.56 gives

$$\Pr[\text{Condorcet winner}] = \frac{3}{4} - \frac{3}{4}\mathrm{Stab}_{-1/3}[f].$$

Also, noise stability has the Fourier form

$$\mathrm{Stab}_\rho[f] = \sum_S \rho^{|S|}\hat{f}(S)^2.$$

CMU School of C... +1

Now take $f = \mathrm{Maj}_3$. Its multilinear expansion is

$$\mathrm{Maj}_3(x_1, x_2, x_3) = \frac{1}{2}(x_1 + x_2 + x_3 - x_1x_2x_3),$$

so its nonzero Fourier coefficients are

$$\hat{f}(\{1\}) = \hat{f}(\{2\}) = \hat{f}(\{3\}) = \frac{1}{2}, \qquad \hat{f}(\{1,2,3\}) = -\frac{1}{2}.$$

Hence

$$W^1[f] = 3 \cdot \left(\frac{1}{2}\right)^2 = \frac{3}{4}, \qquad W^3[f] = \left(-\frac{1}{2}\right)^2 = \frac{1}{4}.$$

Therefore

$$\mathrm{Stab}_\rho[\mathrm{Maj}_3] = \frac{3}{4}\rho + \frac{1}{4}\rho^3.$$

At $\rho = -\frac{1}{3}$,

$$\mathrm{Stab}_{-1/3}[\mathrm{Maj}_3] = \frac{3}{4}\left(-\frac{1}{3}\right) + \frac{1}{4}\left(-\frac{1}{27}\right) = -\frac{1}{4} - \frac{1}{108} = -\frac{7}{27}.$$

Plugging this into Theorem 2.56,

$$\Pr[\text{Condorcet winner}] = \frac{3}{4} - \frac{3}{4}\left(-\frac{7}{27}\right) = \frac{3}{4} + \frac{7}{36} = \frac{27+7}{36} = \frac{34}{36} = \frac{17}{18}.$$

So the precise probability is

$$\boxed{\frac{17}{18}}.$$

This matches the theorem's Condorcet-winner formula specialized to $f = \mathrm{Maj}_3$.

CMU School of C... +1

2.51(a)

Let the three pairwise elections use unanimous rules

$$f,\ g,\ h : \{-1,1\}^n \to \{-1,1\}.$$

As in the proof of Theorem 2.56, a profile of votes is represented by triples $(x_i, y_i, z_i) \in \{-1,1\}^3$ that are coordinatewise in NAE_3, and there is a Condorcet winner exactly when

$$NAE_3(f(x), g(y), h(z)) = 1.$$

CMU School of C...

First show

$$g(x) = -f(-x) \qquad \text{for all } x.$$

Fix any $x \in \{-1,1\}^n$, and consider

$$y = -x, \qquad z = (f(x), \dots, f(x)).$$

This is a valid voting profile: for each coordinate i, the triple

$$(x_i, -x_i, f(x))$$

is never all equal, so it satisfies the NAE_3 condition. Because h is unanimous,

$$h(z) = f(x).$$

Since the election always has a Condorcet winner, the output triple

$$(f(x),\ g(-x),\ h(z)) = (f(x),\ g(-x),\ f(x))$$

must also satisfy NAE_3. The first and third entries are equal, so the middle one must be the opposite sign:

$$g(-x) = -f(x).$$

Replacing x by $-x$ gives

$$g(x) = -f(-x).$$

By cyclic symmetry, the same argument gives

$$h(x) = -g(-x), \qquad f(x) = -h(-x).$$

Combining these,

$$h(x) = -g(-x) = f(x),$$

so $h = f$. Then

$$f(x) = -h(-x) = -f(-x),$$

so f is odd. Finally,

$$g(x) = -f(-x) = f(x).$$

Thus

$$\boxed{f = g = h.}$$

So the "more general" 3-candidate version collapses back to the same rule in all three pairwise elections. The book notes this strengthened formulation in Exercise 2.51 and proves the ordinary 3-candidate Arrow theorem just above it. CMU School of C... +1

2.51(b)

Now consider a Condorcet election with $m > 3$ candidates. For each ordered pair a, b, let F_{ab} be the unanimous Boolean rule used for the pairwise election between a and b.

Take any three candidates a, b, c. Any profile on just these three candidates can be extended to a profile on all m candidates by placing the other $m - 3$ candidates identically for every voter, so the pairwise outcomes among a, b, c are unchanged. Since the full election always has a Condorcet winner, the restricted 3-candidate election can never contain a cycle. Therefore, by part (a),

$$F_{ab} = F_{bc} = F_{ca}.$$

And then by the 3-candidate Arrow theorem, this common rule must be a dictatorship.

CMU School of C...

So for every triple of candidates, all three pairwise rules in that triple are the same dictator. In particular, if two pairs share a candidate, their rules agree. For example, from the triple $\{a, b, c\}$,

$$F_{ab} = F_{ac}.$$

Since the graph of candidate-pairs is connected through overlapping triples, it follows that **all** pairwise rules are the same dictator x_i for one fixed voter i.

Hence for every pair of candidates, society ranks them according to voter i's preference. Therefore the whole social ranking is exactly voter i's ranking. So Arrow's theorem for more than 3 candidates is:

$$\boxed{\text{If a unanimous IIA Condorcet rule always yields a Condorcet winner, then it is a}}$$

If you want, I can also write these in a shorter "submit-this" format.

Sources

.40 Suppose A learns $\mathscr{C}$ from random examples with error $\epsilon/2$ in time T – with probability at least 9/10.

(a) After producing hypothesis h on target $f : \{-1,1\}^n \to \{-1,1\}$, show that A can "check" whether h is a good hypothesis in time $\mathrm{poly}(n, T, 1/\epsilon) \cdot \log(1/\delta)$. Specifically, except with probability at most δ, A should output 'YES' if $\mathrm{dist}(f,h) \le \epsilon/2$ and 'NO' if $\mathrm{dist}(f,h) > \epsilon$. (Hint: Time poly($T$) may be required for A to evaluate $h(x)$.)

(b) Show that for any $\delta \in (0, 1/2]$, there is a learning algorithm that learns $\mathscr{C}$ with error ϵ in time $\mathrm{poly}(n, T, \epsilon) \cdot \log(1/\delta)$ – with probability at least $1 - \delta$.

Solve part (ii)

Thought for 5m 49s >

Let $V(h, \eta)$ be the verifier from part (a): given a hypothesis h, it runs in time $\mathrm{poly}(n, T, 1/\epsilon)\log(1/\eta)$, and with failure probability at most η it satisfies

$$\text{if } \mathrm{dist}(f,h) \le \epsilon/2, \text{ output YES;} \qquad \text{if } \mathrm{dist}(f,h) > \epsilon, \text{ output NO.}$$

The setup of Exercise 3.40 is exactly that A learns with error $\epsilon/2$ in time T with success probability at least $9/10$, and part (a) asks for precisely this kind of verifier.
CMU School of C...

Now do the standard amplification.

Choose

$$m = C\log(1/\delta)$$

for a large enough absolute constant C, and set

$$\eta = \frac{\delta}{2m}.$$

For $t = 1, \dots, m$:

1. Run A independently to get a hypothesis h_t.
2. Run $V(h_t, \eta)$.
3. If the verifier says YES, output h_t and halt.

If no round says YES, output FAIL.

Now analyze this.

Let B be the event that some hypothesis with

$$\mathrm{dist}(f, h_t) > \epsilon$$

is nevertheless accepted by the verifier. By the verifier guarantee and a union bound,

$$\Pr[B] \le m\eta = \delta/2.$$

Next, in any given round, the probability that A produces a genuinely good hypothesis is at least $9/10$. Conditional on that, the verifier says YES except with probability at most η. Hence a given round produces an accepted good hypothesis with probability at least

$$\frac{9}{10}(1-\eta).$$

Since $\delta \le 1/2$, for large enough C we have $\eta \le 1/10$, so this is at least $4/5$.

Therefore the probability that none of the m rounds accepts a good hypothesis is at most

$$(1-4/5)^m = (1/5)^m \le \delta/2$$

for large enough C.

So, with probability at least

$$1 - \delta/2 - \delta/2 = 1 - \delta,$$

both of the following happen:

- no bad hypothesis ($\mathrm{dist} > \epsilon$) is ever accepted;
- at least one good hypothesis ($\mathrm{dist} \le \epsilon/2$) is accepted.

On this event, the algorithm outputs some accepted hypothesis, and since no bad hypothesis is accepted, the output must satisfy

$$\mathrm{dist}(f, h) \le \epsilon.$$

Thus we have amplified the success probability to $1 - \delta$. This is exactly what part (b) asks for. CMU School of C...

The running time is

$$m \cdot \Big(T + \mathrm{poly}(n, T, 1/\epsilon)\log(1/\eta)\Big),$$

which is $\mathrm{poly}(n, T, 1/\epsilon)\log(1/\delta)$ at the level of the exercise's asymptotic statement. So the class is learnable with error ϵ and confidence $1 - \delta$.

Sources

3.41 (a) Our description of the Low-Degree Algorithm with degree k and error ϵ involved using a new batch of random examples to estimate each low-degree Fourier coefficient. Show that one can instead simply draw a single batch $\mathscr{E}$ of $\mathrm{poly}(n^k, 1/\epsilon)$ examples and use $\mathscr{E}$ to estimate each of the low-degree coefficients.

(b) Show that when using the above form of the Low-Degree Algorithm, the final hypothesis $h : \{-1,1\}^n \to \{-1,1\}$ is of the form

$$h(y) = \mathrm{sgn}\left(\sum_{(x, f(x)) \in \mathscr{E}} w(\Delta(y,x)) \cdot f(x)\right),$$

for some function $w : \{0, 1, \dots, n\} \to \mathbb{R}$. In other words, the hypothesis on a given y is equal to a weighted vote over all examples seen, where an example's weight depends only on its Hamming distance to y. Simplify your expression for w as much as you can.

Thought for 2m 48s >

Let

$$N = \sum_{j=0}^{k} \binom{n}{j},$$

the number of Fourier coefficients of degree at most k. In the Low-Degree Algorithm, Theorem 3.29 estimates the coefficients on a set F and then outputs

$$h = \mathrm{sgn}(g), \qquad g = \sum_{S \in F} \tilde{f}(S)\chi_S,$$

using the fact that if every estimate is accurate enough, then $\|f - g\|_2^2 \le \epsilon$, which implies $\mathrm{dist}(f, h) \le \epsilon$. In the low-degree case, $F = \{S : |S| \le k\}$, so $|F| = N = O(n^k)$. CMU School of C... +1

For **(a)**, draw one batch

$$\mathcal{E} = \{(x^{(1)}, f(x^{(1)})), \dots, (x^{(m)}, f(x^{(m)}))\}$$

of i.i.d. random examples, and for every S with $|S| \le k$ define the empirical estimate

$$\tilde{f}(S) = \frac{1}{m}\sum_{t=1}^{m} f(x^{(t)})\chi_S(x^{(t)}).$$

For each fixed S, this is the average of m i.i.d. $\{\pm 1\}$-valued random variables with mean $\hat{f}(S)$, exactly as in Proposition 3.30. So by Hoeffding/Chernoff,

$$\Pr\big[|\tilde{f}(S) - \hat{f}(S)| > \tau\big] \le 2e^{-cm\tau^2}$$

for an absolute constant $c > 0$. Proposition 3.30 is the book's single-coefficient estimate; here we are just reusing the same samples for all S. CMU School of C...

Choose

$$\tau = \frac{\sqrt{\epsilon}}{2\sqrt{N}}.$$

Then if

$$m = O\left(\frac{N \log N}{\epsilon}\right),$$

the union bound gives that with probability at least, say, $9/10$, all N estimates satisfy

$$|\tilde{f}(S) - \hat{f}(S)| \le \tau \qquad \text{simultaneously for all } |S| \le k.$$

Now define

$$g = \sum_{|S| \le k} \tilde{f}(S)\chi_S.$$

Exactly as in the proof of Theorem 3.29,

$$\|f - g\|_2^2 = \sum_{|S| \le k} (\hat{f}(S) - \tilde{f}(S))^2 + \sum_{|S| > k} \hat{f}(S)^2.$$

The first term is at most $N\tau^2 = \epsilon/4$, and the second is at most $\epsilon/2$ by the low-degree concentration assumption. Hence $\|f - g\|_2^2 \le 3\epsilon/4 < \epsilon$, so $h = \mathrm{sgn}(g)$ has error at most ϵ. Thus one batch of size $m = \mathrm{poly}(n^k, 1/\epsilon)$ suffices. CMU School of C... +1

For **(b)**, write the sample as $\mathcal{E} = \{(x, f(x))\}$ and use the same empirical estimates:

$$\tilde{f}(S) = \frac{1}{m}\sum_{(x, f(x)) \in \mathcal{E}} f(x)\chi_S(x), \qquad m := |\mathcal{E}|.$$

Then

$$g(y) = \sum_{|S| \le k} \tilde{f}(S)\chi_S(y) = \frac{1}{m}\sum_{(x, f(x)) \in \mathcal{E}} f(x) \sum_{|S| \le k} \chi_S(x)\chi_S(y).$$

Since $\chi_S(x)\chi_S(y) = \chi_S(xy)$, the inner sum depends only on the vector $xy \in \{-1,1\}^n$, hence only on the number of coordinates where x and y differ, i.e. on $\Delta(x, y)$. This is exactly the form the exercise asks you to show. CMU School of C... +1

So define

$$w(d) := \frac{1}{m} \sum_{|S| \le k} \chi_S(z),$$

where $z \in \{-1,1\}^n$ is any point with exactly d coordinates equal to -1. Then

$$h(y) = \operatorname{sgn}\left(\sum_{(x,f(x)) \in \mathcal{E}} w(\Delta(y,x))\, f(x) \right).$$

An explicit formula for $w(d)$ is obtained by grouping subsets S by size $j = |S|$: if $\Delta(x,y) = d$, then among the j chosen coordinates, suppose t come from the d disagreeing coordinates and $j - t$ come from the $n - d$ agreeing coordinates. Their contribution is $(-1)^t$, and there are $\binom{d}{t}\binom{n-d}{j-t}$ such S. Hence

$$w(d) = \frac{1}{m} \sum_{j=0}^{k} \sum_{t=0}^{j} (-1)^t \binom{d}{t} \binom{n-d}{j-t}.$$

Therefore the final hypothesis is a weighted vote over the examples, with weight depending only on Hamming distance, namely

$$\boxed{h(y) = \operatorname{sgn}\left(\sum_{(x,f(x)) \in \mathcal{E}} \left[\frac{1}{m} \sum_{j=0}^{k} \sum_{t=0}^{j} (-1)^t \binom{\Delta(y,x)}{t} \binom{n - \Delta(y,x)}{j-t} \right] f(x) \right).}$$

That is the desired expression for w. CMU School of C... +1

If you want, I can also rewrite this as a shorter polished homework submission.

Sources

Thought for 1m 19s

Yes—the Goldreich–Levin algorithm extends essentially unchanged to targets

$$f : \{-1,1\}^n \to [-1,1].$$

The original theorem is stated for Boolean-valued f, and its proof relies on two ingredients: estimating a single Fourier coefficient $\hat{f}(U)$, and estimating bucket weights

$$W_{S|J}[f] = \sum_{T \subseteq J} \hat{f}(S \cup T)^2,$$

using the identity

$$W_{S|J}[f] = \mathbb{E}_z\left[\widehat{f_{J|z}}(S)^2\right].$$

Those are Proposition 3.30, Definition 3.39, Corollary 3.22, and Proposition 3.40 in the book. CMU School of C... +2

The point is that nothing in the divide-and-conquer part of the proof actually needs $f \in \{-1,1\}$; it only needs the quantities being sampled to be bounded, so that they can be estimated from random samples in polynomial time. For $f : \{-1,1\}^n \to [-1,1]$, this remains true. CMU School of C... +1

First, for any fixed $U \subseteq [n]$,

$$\hat{f}(U) = \mathbb{E}_x[f(x)\chi_U(x)].$$

Now $\chi_U(x) \in \{-1,1\}$ and $f(x) \in [-1,1]$, so

$$X := f(x)\chi_U(x) \in [-1,1].$$

Thus $\hat{f}(U)$ is the mean of a bounded random variable. By Hoeffding's inequality, from

$$m = O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$$

random examples one can estimate $\hat{f}(U)$ to within $\pm\varepsilon$ with failure probability at most δ. This is the exact analogue of Proposition 3.30, with Hoeffding replacing the $\{\pm 1\}$-valued Chernoff estimate used there. CMU School of C...

Second, for bucket weights, the proof of Proposition 3.40 rewrites

$$W_{S|J}[f] = \mathbb{E}_{z,y,y'}[f(y,z)\chi_S(y)\, f(y',z)\chi_S(y')].$$

When $f \in [-1,1]$, the sampled quantity

$$Y := f(y,z)\chi_S(y)\, f(y',z)\chi_S(y')$$

still lies in $[-1,1]$, because it is a product of two numbers in $[-1,1]$ and two signs. Hence $W_{S|J}[f]$ can again be estimated to within $\pm\varepsilon$ using

$$O\left(\frac{\log(1/\delta)}{\varepsilon^2}\right)$$

samples, in polynomial time. So Proposition 3.40 also extends verbatim. CMU School of C...

Once these two estimation subroutines are available, the rest of the Goldreich–Levin proof is unchanged: we recursively split buckets, estimate the corresponding $W_{S|J}[f]$'s, discard buckets whose estimated weight is too small, and keep subdividing the others until singletons are reached. The correctness argument depends only on accurate estimation of these weights and on Parseval-style control of total Fourier weight, not on Booleanity of f. Therefore we get the same conclusion: given query access to $f : \{-1,1\}^n \to [-1,1]$ and $0 < \tau \le 1$, there is a $\mathrm{poly}(n, 1/\tau)$-time algorithm that, with high probability, outputs a list L such that

$$|\hat{f}(U)| \ge \tau \implies U \in L, \qquad U \in L \implies |\hat{f}(U)| \ge \tau/2.$$

This is exactly the Boolean theorem's guarantee, now for bounded real-valued targets. CMU School of C...

The only extra comment needed for the exercise's model is computational: since every queried value $f(x)$ is promised to be a rational with $\mathrm{poly}(n)$ bits, each sample and each arithmetic operation used in the estimators can be carried out in polynomial time, so the runtime bound remains polynomial. That promise is part of Exercise 3.42 itself. CMU School of C...

Finally, the output list is still small. By Parseval,

$$\sum_U \hat{f}(U)^2 = \mathbb{E}[f(x)^2] \le 1$$

because $f(x) \in [-1,1]$. Hence if every reported U satisfies $|\hat{f}(U)| \ge \tau/2$, then

$$|L| \cdot (\tau/2)^2 \le 1,$$

so

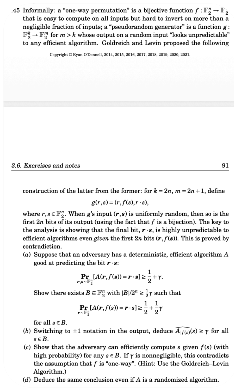
$$|L| \le \frac{4}{\tau^2}.$$

Thus the quantitative part of the theorem is unchanged as well. CMU School of C...

A compact writeup is:

> Replace every use of "the sampled random variable is ± 1-valued" in Propositions 3.30 and 3.40 by "the sampled random variable lies in $[-1,1]$", and apply Hoeffding's inequality instead of the $\{\pm 1\}$-Chernoff bound. Then $\hat{f}(U)$ and $W_{S|J}[f]$ are still estimable in $\mathrm{poly}(n, 1/\varepsilon)\log(1/\delta)$ time. The recursive bucket-splitting proof of Goldreich–Levin is otherwise identical, so the theorem holds verbatim for $f : \{-1,1\}^n \to [-1,1]$. Since $\sum_U \hat{f}(U)^2 = \mathbb{E}[f^2] \le 1$, the output list still has size at most $4/\tau^2$.

Sources


.45 Informally: a "one-way permutation" is a bijective function $f : \mathbb{F}_2^n \to \mathbb{F}_2^n$ that is easy to compute on all inputs but hard to invert on more than a negligible fraction of inputs; a "pseudorandom generator" is a function $g : \mathbb{F}_2^k \to \mathbb{F}_2^m$ for $m > k$ whose output on a random input "looks unpredictable" to any efficient algorithm. Goldreich and Levin proposed the following

construction of the latter from the former: for $k = 2n$, $m = 2n+1$, define

$$g(r,s) = (r, f(s), r \cdot s),$$

where $r, s \in \mathbb{F}_2^n$. When g's input $(\boldsymbol{r}, \boldsymbol{s})$ is uniformly random, then so is the first $2n$ bits of its output (using the fact that f is a bijection). The key to the analysis is showing that the final bit, $\boldsymbol{r} \cdot \boldsymbol{s}$, is highly unpredictable to efficient algorithms even *given* the first $2n$ bits $(\boldsymbol{r}, f(\boldsymbol{s}))$. This is proved by contradiction.

(a) Suppose that an adversary has a deterministic, efficient algorithm A good at predicting the bit $\boldsymbol{r} \cdot \boldsymbol{s}$:
$$\Pr_{\boldsymbol{r},\boldsymbol{s} \sim \mathbb{F}_2^n}[A(\boldsymbol{r}, f(\boldsymbol{s})) = \boldsymbol{r} \cdot \boldsymbol{s}] \ge \frac{1}{2} + \gamma.$$
Show there exists $B \subseteq \mathbb{F}_2^n$ with $|B|/2^n \ge \frac{1}{2}\gamma$ such that
$$\Pr_{\boldsymbol{r} \sim \mathbb{F}_2^n}[A(\boldsymbol{r}, f(s)) = \boldsymbol{r} \cdot s] \ge \frac{1}{2} + \frac{1}{2}\gamma$$
for all $s \in B$.

(b) Switching to ± 1 notation in the output, deduce $\widehat{A_{|f(s)}}(s) \ge \gamma$ for all $s \in B$.

(c) Show that the adversary can efficiently compute s given $f(s)$ (with high probability) for any $s \in B$. If γ is nonnegligible, this contradicts the assumption that f is "one-way". (Hint: Use the Goldreich–Levin Algorithm.)

(d) Deduce the same conclusion even if A is a randomized algorithm.

Thought for 2m 52s

Let

$$p_s := \Pr_{r \sim \mathbb{F}_2^n}[A(r, f(s)) = r \cdot s].$$

The hypothesis is

$$\mathbb{E}_s[p_s] = \Pr_{r,s}[A(r, f(s)) = r \cdot s] \ge \frac{1}{2} + \gamma.$$

The exercise asks you to turn this predictor into an inverter, using the Goldreich–Levin algorithm for finding large Fourier coefficients. CMU School of C... +1

(a)

Define

$$B := \left\{ s \in \mathbb{F}_2^n : \ p_s \ge \frac{1}{2} + \frac{\gamma}{2} \right\}.$$

We show $|B|/2^n \ge \gamma/2$.

Suppose not, and write $\alpha := |B|/2^n < \gamma/2$. Since always $p_s \le 1$, we would have

$$\mathbb{E}_s[p_s] \le \alpha \cdot 1 + (1-\alpha)\left(\frac{1}{2} + \frac{\gamma}{2}\right) = \frac{1}{2} + \frac{\gamma}{2} + \alpha\frac{1-\gamma}{2}.$$

Using $\alpha < \gamma/2$,

$$\mathbb{E}_s[p_s] < \frac{1}{2} + \frac{\gamma}{2} + \frac{\gamma}{2} \cdot \frac{1-\gamma}{2} = \frac{1}{2} + \frac{3\gamma}{4} - \frac{\gamma^2}{4} < \frac{1}{2} + \gamma,$$

contradicting the hypothesis. Hence

$$\boxed{\frac{|B|}{2^n} \geq \frac{\gamma}{2}.}$$

So on at least a $\gamma/2$ fraction of s's, the predictor has conditional advantage at least $\gamma/2$. CMU School of C...

(b)

Now switch the output notation to $\{\pm 1\}$. For each fixed $y \in \mathbb{F}_2^n$, define

$$A_{|y}(r) := (-1)^{A(r,y)} \in \{\pm 1\}.$$

Also recall the Fourier characters on $\mathbb{F}_2^n$:

$$\chi_s(r) = (-1)^{r \cdot s}.$$

For any fixed s,

$$\Pr_r[A(r, f(s)) = r \cdot s] = \Pr_r\left[A_{|f(s)}(r) = \chi_s(r)\right].$$

For $\{\pm 1\}$-valued random variables X, Y,

$$\Pr[X = Y] = \frac{1}{2}\big(1 + \mathbb{E}[XY]\big).$$

Therefore

$$p_s = \frac{1}{2}\left(1 + \mathbb{E}_r\left[A_{|f(s)}(r)\chi_s(r)\right]\right) = \frac{1}{2}\left(1 + \widehat{A_{|f(s)}}(s)\right).$$

If $s \in B$, then $p_s \geq \frac{1}{2} + \frac{\gamma}{2}$, so

$$\widehat{A_{|f(s)}}(s) \geq \gamma.$$

Thus

$$\boxed{\widehat{A_{|f(s)}}(s) \geq \gamma \quad \text{for all } s \in B.}$$

This is exactly the "large Fourier coefficient" conclusion needed for Goldreich–Levin. CMU School of C... +1

(c)

Fix $s \in B$, and suppose we are given $y = f(s)$. Consider the function

$$g_y(r) := A_{|y}(r).$$

Because A is efficient, we have efficient query access to g_y. By part (b),

$$\widehat{g_y}(s) = \widehat{A_{|f(s)}}(s) \geq \gamma.$$

Now run the Goldreich–Levin algorithm on g_y with threshold $\tau = \gamma$. With high probability it outputs a list L containing every U with $|\widehat{g_y}(U)| \geq \gamma$, and every listed U satisfies $|\widehat{g_y}(U)| \geq \gamma/2$; moreover $|L| \leq 4/\gamma^2$. So $s \in L$ with high probability. CMU School of C...

Finally, since f is an efficiently computable permutation, we can compute $f(u)$ for every $u \in L$ and output the unique u such that

$$f(u) = y.$$

Because f is bijective, that unique u is exactly s. Hence we can efficiently recover s from $f(s)$ for every $s \in B$, with high probability. Since part (a) gives

$$\Pr[s \in B] \geq \gamma/2,$$

if γ is nonnegligible then this inverts f on a nonnegligible fraction of inputs, contradicting that f is one-way. CMU School of C... +1

(d)

Now suppose A is randomized. Let ω denote its internal random tape, and let A_ω be the deterministic algorithm obtained by fixing ω. The success probability of A is

$$\mathbb{E}_\omega \Pr_{r,s}[A_\omega(r, f(s)) = r \cdot s].$$

If this average is at least $\frac{1}{2} + \gamma$, then for some fixing ω^*,

$$\Pr_{r,s}[A_{\omega^*}(r, f(s)) = r \cdot s] \geq \frac{1}{2} + \gamma.$$

So some deterministic version A_{ω^*} has the same advantage. Hardwire ω^* into the algorithm and apply parts (a)–(c). This again yields an efficient inverter for f on a nonnegligible fraction of inputs, a contradiction. Therefore the same conclusion holds even when A is randomized. CMU School of C... +1

So the whole exercise proves the Goldreich–Levin cryptographic reduction:

$$\text{if } (r, f(s)) \text{ lets you predict } r \cdot s \text{ with nonnegligible advantage, then } f \text{ is invertible}$$

Sources

ChatGPT